Wavelet Tree Transformer: Multihead Attention With Frequency-Selective Representation and Interaction for Remote Sensing Object Detection

Jiahao Pan, Chu He, *Senior Member, IEEE*, Wei Huang, Jidong Cao, and Ming Tong

***Abstract*— Vision Transformer has achieved remarkable success in image recognition tasks owing to its global modeling ability. However, the quadratic computational complexity becomes a prominent issue when dealing with high-resolution remote sensing images. Numerous studies have explored the potential of spectral analysis to reveal for more discriminative features. However, neural network exhibit frequency tendency, and different features are interested in different frequencies. Unfortunately, there is no well-established criterion for selecting appropriate frequency representations. To address these issues, a novel wavelet tree head attention (WTHA-ViT) model is proposed which combines a tree structure on the wavelet frequencies with multihead attention in the Transformer encoder, possessing the ability to interact with cross-combinations of short and long-range as well as high and low-frequency components. First, we construct a wavelet tree reduction module (WTRM) based on the wavelet tree structure, utilizing the wavelet decomposition to retain frequency features suitable for each patch, which enables global modeling with various frequency components while reducing computational complexity. Second, guided by channel correlations, we propose the channel lifting scheme multihead attention (CLSMHA) to model the importance on the heads of multihead attention and focus on the more salient head features. Finally, our WTHA-ViT can replace the backbone of detection networks for dense prediction tasks. Extensive experiments on DOTA-V1.0 and HRSID datasets demonstrate that our model exhibits superior performance and robustness compared to state-of-the-art networks. Besides, we evaluate the transferability of the model on DIOR and LEVIR datasets and verify its generalization ability. The code is available at https://github.com/conquer-pan/WTHA-ViT.**

***Index Terms*— Lifting scheme, multihead, object detection, remote sensing, transformer, wavelet tree structure.**

I. Introduction

OBJECT detection is a fundamental task in computer vision that aims to locate and identify objects of interest [1], [2], [3]. Thanks to the substantial development of remote sensing imaging technology and the availability of abundant remote sensing data, it has drawn significant attention to object detection of remote sensing images in recent years [4], [5]. This interest is driven by its crucial implications and applicability in various domains such as military, agriculture, and disaster monitoring [6]. However, contemporary applications of remote sensing data still face formidable challenges [7], including complex backgrounds and higher resolutions. Specifically, remote sensing images frequently contain diverse objects like buildings, roads, and vegetation, which introduce a wide array of shape, texture, and color characteristics, adding complexity to the background. Moreover, in synthetic aperture radar (SAR) images [8], [9], with their unique imaging mechanisms, objects are prone to being mistaken for background elements. Additionally, high-resolution remote sensing images impose a significant computational burden on existing mainstream object detection models [10]. Therefore, effectively addressing these challenges has become a crucial aspect of remote sensing object detection.

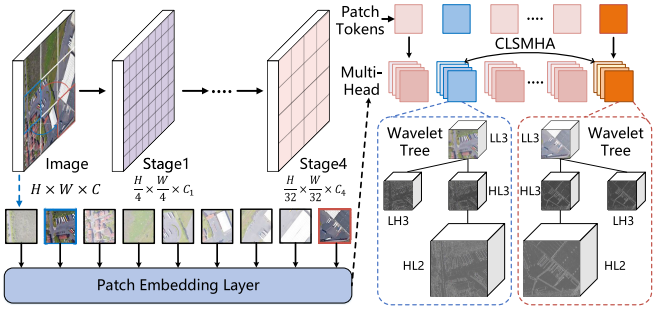

Fig. 1. Our model utilizes the wavelet decomposition tree structure to retain the frequency features appropriate for each patch for global modeling.

To detect objects accurately in remote sensing images with complex backgrounds, models need strong feature extraction and global context modeling abilities. Recently, the Transformer [11] has shown remarkable success in computer vision, revolutionizing various downstream tasks [12], [13], [14], [15], [16]. The Transformer's encoder extracts feature representations of the input sequence and captures context information. In the decoder [17], the cross-attention unit establishes associations between the decoder object query and input features, exploring similarity between the decoder query

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and encoder features. Besides, the Vision Transformer [14] represents the earliest exploration of self-attention, allowing information interaction between each visual token and capturing long-range relationships between patches, which is lacking in convolutional neural networks (CNNs).

However, Transformer suffers from a quadratic computational complexity with respect to the input size [14], especially in the case of high-resolution remote sensing images. To address this issue, there are currently three main directions for solutions: 1) some studies have explored the sparsity of the attention matrix [18], [19], [20], significantly reducing computational complexity. However, it may have an impact on the overall performance; 2) Swin [13] and Twins [21] models introduced the concept of using a shifted window instead of a fixed window. This window-based attention method effectively reduces computational requirements while preserving global dependencies, but it may lead to information loss at the window boundaries; and 3) the most commonly used method involves downsampling the key and value features in self-attention. PVT [22] performs downsampling using simple convolution or pooling operations while WaveViT [23] adopts wavelet transform. Before the wavelet decomposition, it reduces the number of channels to a quarter of the original. These methods will cause a certain degree of information loss.

Furthermore, recent studies have demonstrated the discriminative image features revealed through frequency domain analysis [24], [25]. SpectFormer [26] acknowledges the significance of both spectral and multiattention approaches. GFNet [27] shows that the Fourier transform can replace the multihead attention layer while maintaining comparable performance. Building upon this research, the AFNO [28] introduces token mixers as adaptive Fourier neural operators. Wavevit [23] utilizes wavelet transform for downsampling learning—an effective time-frequency analysis tool that decomposes signals into various frequency subbands to address aliasing issues. Wavelet transform [29] has been widely used to complement the multifrequency information and facilitate downsampling learning. These studies underscore the advantages of frequency domain analysis in Transformers. Nevertheless, several works have revealed the frequency biases in neural networks [30], indicating varying interested frequency features across different feature maps. Within the mainstream Transformer architecture, most methods [23], [31], [32] tend to employ all frequency features indiscriminately for global modeling. However, there is not a well-defined criterion for selecting appropriate frequency components based on each patch's characteristics for interaction. Additionally, existing methods do not fully filter out the noise in high-frequency features, preserving only the texture information.

Based on the above analysis, we propose a novel and effective Transformer encoder structure called wavelet tree head attention (WTHA-ViT), which can replace the backbone in the object detection network for the prediction task. WTHA-ViT fully leverages wavelets to innovate frequency attention. It employs a tree structure along the wavelet decomposition to determine the selection of multiple frequency components, reducing computational complexity and feature redundancy. As shown in Fig. 1, context modeling relies on frequency features suitable for each object, enabling different frequency components of patches to be combined to achieve long-range dependencies. They are then aggregated at the head to model the importance of different combinations. In detail, our proposed wavelet tree reduction module (WTRM) utilizes the downsampling function of the wavelet transform to construct a spatial reduction module based on the wavelet tree structure [33], [34]. It uses a wavelet tree selection (WTS) strategy to retain the most interesting frequency features, which allows the model to learn the most salient features for each object, addressing the issue of losing object structure and texture detail information caused by indiscriminate removal of certain frequency components. The channel lifting scheme multihead attention (CLSMHA) module we designed considers the correlation between channels and constructs high-frequency and low-frequency features along the channel distribution, which utilizes the second-generation wavelet transform [35], [36] to model the heads of the Transformer, enabling the network to focus on the more important head features. Extensive experiments have been conducted on two benchmark datasets, namely, the remote sensing optical dataset DOTA-V1.0 [37] and the SAR dataset HRSID [38] to validate its effectiveness and robustness. Overall, the contributions are summarized as follows:

1) A novel WTHA-ViT is introduced to leverage wavelet transform to enhance frequency attention, which utilizes a tree structure based on wavelet decomposition to select frequency representation. Combined with the Transformer architecture, it facilitates the integration of different frequency components for both long- and short-range correlations.
2) Two effective modules are proposed to enhance Transformer architecture interaction: WTRM reduces computational complexity while selecting suitable frequency components for each patch by a WTS strategy. CLSMHA performs head importance modeling through second-generation wavelets to focus on the most salient head features.
3) Extensive experiments on four datasets to validate the effectiveness and robustness of WTHA-ViT. Additionally, our model outperforms current state-of-the-art models.

II. Related Work

A. Methods to Reduce Complexity in Transformer

For a feature map $X \in \mathbb{R}^{(C,H,W)}$, the computational cost of multihead attention is $O(H^2W^2C)$, which results in quadratic complexity for the Transformer. To tackle this challenge, several innovative approaches [13], [18], [19], [20], [21], [22], [23], [41] have been proposed. These methods mainly fall into four categories: sparsifying attention matrices, window attention, alternative to self-attention in token mixers, and downsampling of key and value features.

1) The method of sparse attention matrices reduces computational cost by setting some attention weights to zero

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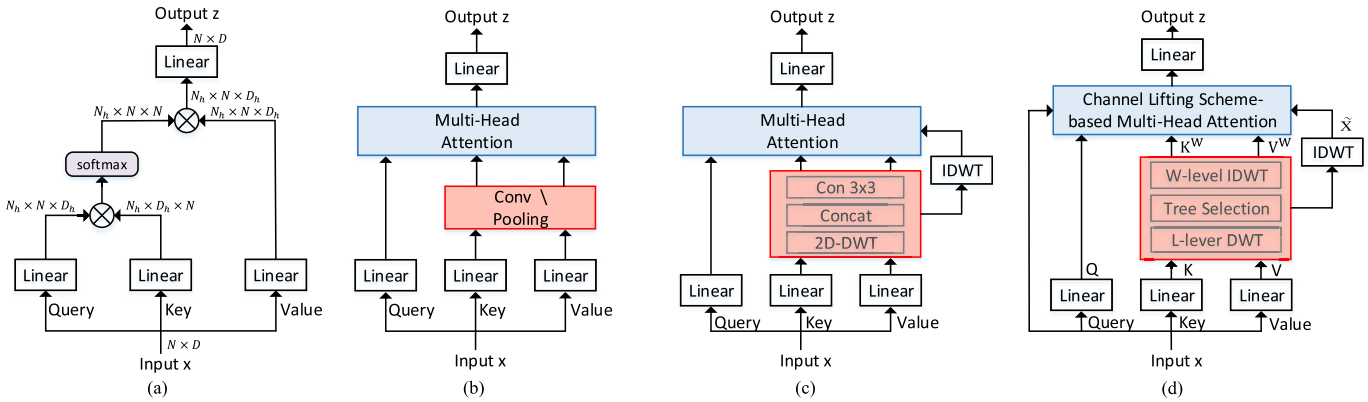

Fig. 2. Multihead attention with different downsample operation. (a) Multihead attention mechanism in ViT [14]. (b) PVT [22] employs downsampling learning on the K and V features through convolution or pooling. (c) Wavevit [23] compresses the number of channels before using a simple wavelet transform for downsampling learning. (d) Our wavelet tree structure selects appropriate frequency features.

and only computing the nonzero values. Reformer [18] utilizes locality-sensitive hashing for attention matrix selection, while Sparse Transformer [19] employs a top-k selection strategy to sparsify the attention. Longformer [20] combines local and global attention, significantly decreasing computational complexity. However, these approaches may also affect performance.

2) The window attention method facilitates interaction between patch windows by transforming fixed windows into shifted windows, such as Swin [13], Twins [21], and CSWin [41]. SepViT [42] enhances the interaction within and between windows by employing depth-wise separable self-attention. This method effectively reduces computational complexity while preserving global dependencies, but it may lead to the loss of information at window boundaries.

3) The method to replace the self-attention in token mixers. Many studies show that a key component of transformer effectiveness is due to the proper mixing of tokens. The MLP-mixer [43] achieves similar accuracy as self-attention. In addition, Fourier-based Mixers have also been developed. FNet [25] performs FFT on the input and directly adds the real part of the spectrum to the tokens. GFNet [27] uses the weights of different channels to perform element-level multiplication between learnable global filters, and realizes log-linear complexity. AFNO [28] as an efficient token mixer that learns to mix in the Fourier domain and it is highly parallel with a quasi-linear complexity. BoTNe [44] uses convolution and self-attention mechanisms to maintain global and local information. It replaces 3×3 convolution with MHSA in the last three bottlenecks of ResNet.

4) The most commonly used approach involves downsampling the key and value features in self-attention. PVT [22] employs convolution or pooling operation, whereas Wavevit [23] reduces the number of channels before applying wavelet transformation and treats all four subbands equally, as depicted in Fig. 2(b) and (c). Nonetheless, these techniques exhibit a restricted range of interaction and may consequently experience information loss. Consequently, our research concentrates on analyzing the wavelet tree structure and selecting suitable frequency features for interaction.

B. Frequency Domain Analysis of Transformer

Given the potential of the frequency domain in capturing rich image features, we explore the spectral analysis within the Transformer framework in Table I. Where $N = HW$, d, k, and r refer to the number of tokens, the number of channels, block count, and reduction factor. Note that for the SpectFormer, we list the entire network and L, α refer to the number of all SpectFormer blocks, the number of spectral blocks.

HAT [39] believes that ViT is inferior to CNNs in capturing high-frequency components of images, and it directly enhances the high-frequency components of images through adversarial training. Prism [40] proposes a prism layer training model via DCT that uses spectral filters to constrain the structure of different neurons at different scales. Spectral gating network (SGN) is proposed to mix tokens from the perspective of frequency which can replace multihead self-attention (MSA) to reduce complexity. FNet [25] performs FFT on the input and directly adds the real part of the spectrum to the token, thus mixing the information from the spatial domain and the frequency domain. GFNet [27] demonstrates that Fourier Transform can replace multihead attention layers while achieving comparable performance. AFNO [28] further extends this research by treating token mixers as an adaptive Fourier neural operator. SpectFormer [26] thinks that SGN proposed by FNet, FGNet, AFNO, and the original MSA play a major role and it proposes a stacked SGN and MSA structure. The specific content differences of Transformer method in frequency domain are shown in Table I.

Wavelet transform is a powerful tool for signal time-frequency analysis. It has been widely applied in the field of CNNs, including tasks such as replacing pooling and convolutional layers with downsampling [36], [45], [46], feature supplementation [47], image decomposition, and reconstruction [48], [49]. Research on wavelet transform-based Transformer is limited, primarily originating from WaveViT [23]. WaveViT performs downsampling on the key and value features. However, WaveViT only utilizes the lowest-frequency feature for the interaction between different patches. Similarly, our method utilizes a wavelet tree-based Transformer via a self-adaptive reduction factor r that selects appropriate frequency features to participate in the computation and

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TABLE I

COMPARISONS OF TRANSFORMER MODELS BASED ON FREQUENCY ANALYSIS. WE COMPARE TRANSFORMER MODELS BASED ON ATTENTION TYPE, TOKEN MIXERS, FREQUENCY TRANSFORM METHOD, REDUCE COMPLEXITY, COMPLEXITY, AND PARAMETERS. SGN STANDS FOR A SGN, AND MSA STANDS FOR A MULTIHEAD SELF ATTENTION NETWORK. "‡" INDICATES THAT IT ADDS A SPECTRUM LAYER BEHIND MSA. GLOBAL STANDS FOR GLOBAL ATTENTION MODELING AND "*" INDICATES THAT IT IS NOT AN ATTENTION-TYPE NETWORK

Model	Attention Type	Token Mixers	Frequency Transform	Reduce Complexity	Complexity	Params
ViT [14]	Global	MSA	None	✗	$\mathcal{O}(N^2d+Nd^2)$	$3d^2$
HAT [39]	Global	MSA	DFT	✗	$\mathcal{O}(N^2d+Nd^2)$	$3d^2$
Prism [40]	Global	MSA‡	DCT	✗	$\mathcal{O}(N^2d+Nd^2)$	$3d^2$
FNet [25]	Global*	SGN	FFT	✓	$\mathcal{O}(Ndlog(N)+Ndlog(d))$	Nd
GFNet [27]	Global*	SGN	FFT	✓	$\mathcal{O}(Ndlog(N)+Nd)$	Nd
AFNO [28]	Global*	SGN	FFT	✓	$\mathcal{O}(Ndlog(N)+\frac{Nd^2}{k})$	$(1+\frac{k}{4})d^2+4d$
WaveViT [23]	Global	MSA	DWT	✓	$\mathcal{O}(\frac{N^2}{r^2}d+Nd^2r^2)$	$3d^2$
WTHA-ViT (ours)	Global	MSA	DWT	✓	$\mathcal{O}(\frac{N^2}{r^2}d+Nd^2)$	$3d^2$
SpectFormer [26]	Global	SGN + MSA	FFT	✓	$\mathcal{O}(\alpha(Ndlog(N)+Nd)$ $+(L-\alpha)(N^2d+Nd^2))$	$\alpha Nd+$ $(L-\alpha)(3d^2)$

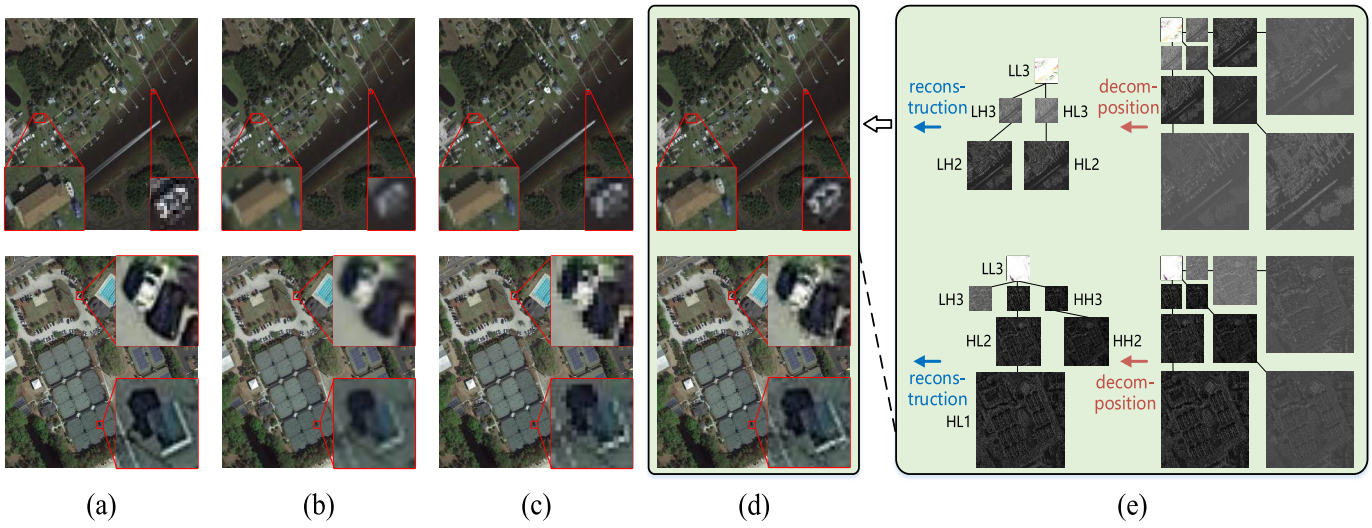

Fig. 3. Comparison of different downsampling and reconstruction methods. (a) Original images. (b) Reconstructed images after pooling downsampling. (c) Reconstructed images by inverse transform using only wavelet low-frequency component LL. (d) Reconstructed image after frequency selection through wavelet tree structure (ours). (e) Detailed processing diagram of our wavelet tree frequency selection strategy.

interaction, thereby maximizing the preservation of feature information.

C. Frequency Component Selection Strategy

Numerous studies have demonstrated the frequency bias of neural networks [26], [30], wherein different frequency features contain distinct information and serve as representations of the feature maps. However, the interested frequency features may not be consistent across different feature maps. Based on current research, the selection of frequency components can be categorized into the following three strategies.

The first strategy involves preserving low-frequency components and discarding high-frequency components [50]. Generally, the low-frequency features contain the main structural information, while the high-frequency features store a significant amount of noise information while preserving the detailed information. This strategy allows for saving computational resources while maintaining excellent performance.

The second strategy [23], [45], [51] entails retaining all frequency components without discrimination and involves fusing all frequency components together. It includes methods like WCAM [51], where multiple frequency features are weighted and fused, and Wang et al.'s approach [45], which treats high-frequency components as attention weights to update the information of low-frequency components.

The final strategy is to select the optimal frequency components. FCANet [30] proposes two methods for extracting the best frequency components: the Two Step method, which evaluates the importance of each frequency component and selects the one with the best performance based on the evaluation results but has the disadvantage of not achieving end-to-end training, and the NAS method, which

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employs a neural structure search algorithm to find the best frequency components but leads to increased training time for the network. Through analysis, it is evident that there is currently no optimal structure for selectively filtering specific frequency features of the object. Therefore, we focus on the differentiated selection of frequency components, aiming to utilize a more effective structure for feature selection.

III. Preliminaries

A. Principle of Wavelet Transform

Performing a 2-D discrete wavelet transform (DWT) on a feature map involves applying 1-D DWTs in the row and column dimensions. According to the 2-D discrete wavelet theory, a 2-D signal $f(x, y)$ can be decomposed into a combination of four functions with different scales and positions as

$$\begin{aligned} f(x,y) &= \frac{1}{\sqrt{MN}}\sum_m\sum_n W_\varphi(j_0,m,n)\varphi_{j_0,m,n}(x,y) \\ &\quad + \frac{1}{\sqrt{MN}}\sum_{j=0}^{\infty}\Bigg\{\sum_m\sum_n W_\psi^h(j,m,n)\psi_{j,m,n}^h(x,y) \\ &\qquad + \sum_m\sum_n W_\psi^v(j,m,n)\psi_{j,m,n}^v(x,y) \\ &\qquad + \sum_m\sum_n W_\psi^d(j,m,n)\psi_{j,m,n}^d(x,y)\Bigg\} \\ &= Af_a + Hf_h + Vf_v + Df_d \end{aligned} \tag{1}$$

where M, N, h, v, d represent the width and height of the signal, the horizontal, vertical, and diagonal directions, respectively. As in the 1-D case, j_0 is an arbitrary starting scale. The approximate coefficient A for scale j_0 and horizontal, vertical, diagonal detailed coefficients H, V, D for scales $j \geq j_0$ are as

$$\begin{aligned} A = W_\varphi(j_0,m,n) &= \frac{1}{\sqrt{MN}}\sum_{x=0}^{M-1}\sum_{y=0}^{N-1} \\ &\quad \times f(x,y)\varphi_{j_0,m,n}(x,y) \end{aligned} \tag{2}$$

$$\begin{aligned} \{H,V,D\} &= W_\psi^i(j,m,n), i = \{h,v,d\} \\ &= \frac{1}{\sqrt{MN}}\sum_{x=0}^{M-1}\sum_{y=0}^{N-1} f(x,y)\psi_{j,m,n}^i(x,y) \end{aligned} \tag{3}$$

We normally let $j_0 = 0$ and select $N = M = 2^J$ so that $j = 0, 1, 2, \ldots, J-1$ and $m = n = 0, 1, 2, \ldots, 2^j - 1$. The scaling function and wavelet function for different scales j and positions m, n are defined as follows:

$$\varphi_{j,m,n}(x,y) = 2^{\frac{j}{2}}\varphi\left(2^j x - m, 2^j y - n\right) \tag{4}$$

$$\psi_{j,m,n}^i(x,y) = 2^{\frac{j}{2}}\psi_{j,m,n}^i\left(2^j x - m, 2^j y - n\right). \tag{5}$$

With a given scaling function $\varphi(x)$ and wavelet function $\psi(y)$, one 2-D scaling function $\varphi(x, y)$ and three 2-D wavelet functions $\psi(x, y)$ can be obtained as

$$\begin{aligned} \varphi(x,y) &= \varphi(x)\varphi(y) \\ \psi^h(x,y) &= \psi(x)\varphi(y) \\ \psi^v(x,y) &= \varphi(x)\psi(y) \\ \psi^d(x,y) &= \psi(x)\psi(y). \end{aligned} \tag{6}$$

In this article, we use discrete Haar wavelet with scaling function and wavelet function as

$$\{\varphi(n)|n = 0,1\} = \left\{\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right\} \tag{7}$$

$$\{\psi(n)|n = 0,1\} = \left\{\frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}}\right\}. \tag{8}$$

Given an image I of size (n, n, c), the 2-D DWT can be achieved by matrices, as [50]

$$\begin{aligned} H &= \begin{bmatrix} h_0 & h_1 & \cdots & h_{n-2} & h_{n-1} \\ h_{n-2} & h_{n-1} & \cdots & h_{n-4} & h_{n-3} \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ h_2 & h_3 & \cdots & h_0 & h_1 \end{bmatrix} \\ G &= \begin{bmatrix} g_0 & g_1 & \cdots & g_{n-2} & g_{n-1} \\ g_{n-2} & g_{n-1} & \cdots & g_{n-4} & g_{n-3} \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ g_2 & g_3 & \cdots & g_0 & g_1 \end{bmatrix} \end{aligned} \tag{9}$$

$$\begin{bmatrix} LL & HL \\ LH & HH \end{bmatrix} = \begin{bmatrix} H \\ G \end{bmatrix} I \begin{bmatrix} H \\ G \end{bmatrix}^{\mathrm{T}} \tag{10}$$

where H and G are the generating matrices of size $((n/2), n, c)$ populated with Haar wavelet basis by (7) and (8). Its basis is formed by $h = ((1/(2)^{1/2}), (1/(2)^{1/2}))$ and $g = ((1/(2)^{1/2}), -(1/(2)^{1/2}))$. After the DWT, the input feature map generates four subbands LL, LH, HL, and HH, each with half the width and height of the original size. These subbands represent the low-frequency main components, vertical high-frequency information, horizontal high-frequency information, and diagonal high-frequency information, respectively.

The wavelet tree structure [52] is a method based on the wavelet transform. Its fundamental principle is to decompose a signal through multiple DWTs into multiple subsignals of different scales and resolutions, which are then organized and processed by a tree structure.

B. Multihead Attention Mechanism in ViT

In Fig. 2(a), in the mainstream Transformer, the multihead attention mechanism takes the features of Patch Embedding and maps them into query, key, and value features through three linear transformations. For a given input feature map $X \in \mathbb{R}^{(H\times W\times C)}$, it is divided into N patches of size P^2C by setting the patch size P, and reshaped into a patch with dimension $N \times (\mathrm{P}^2C)$, where $N = (HW/\mathrm{P}^2)$. Finally, its dimension is transformed into D by linear transformation to obtain the input feature map $X_p \in \mathbb{R}^{(N\times D)}$ for multihead attention.

Then, they are split along the channels and flattened. At this point, the $Q, K, V \in \mathbb{R}^{(N_h\times N\times D_h)}$, where N_h is the number of heads and $D_h = (D/N_h)$ is the dimension of each head. The $Q_j \in \mathbb{R}^{(N\times D_h)}$, $K_j \in \mathbb{R}^{(N\times D_h)}$, and $V_j \in \mathbb{R}^{(N\times D_h)}$ represent the jth head$_j$. For each head

$$\mathrm{head}_j = \mathrm{Softmax}\left(Q_j {K_j}^{\mathrm{T}}/\sqrt{D_h}\right)V_j \tag{11}$$

$$M = \mathrm{Concat}\left(\mathrm{head}_1, \mathrm{head}_2, \ldots, \mathrm{head}_{N_h}\right)W^o \tag{12}$$

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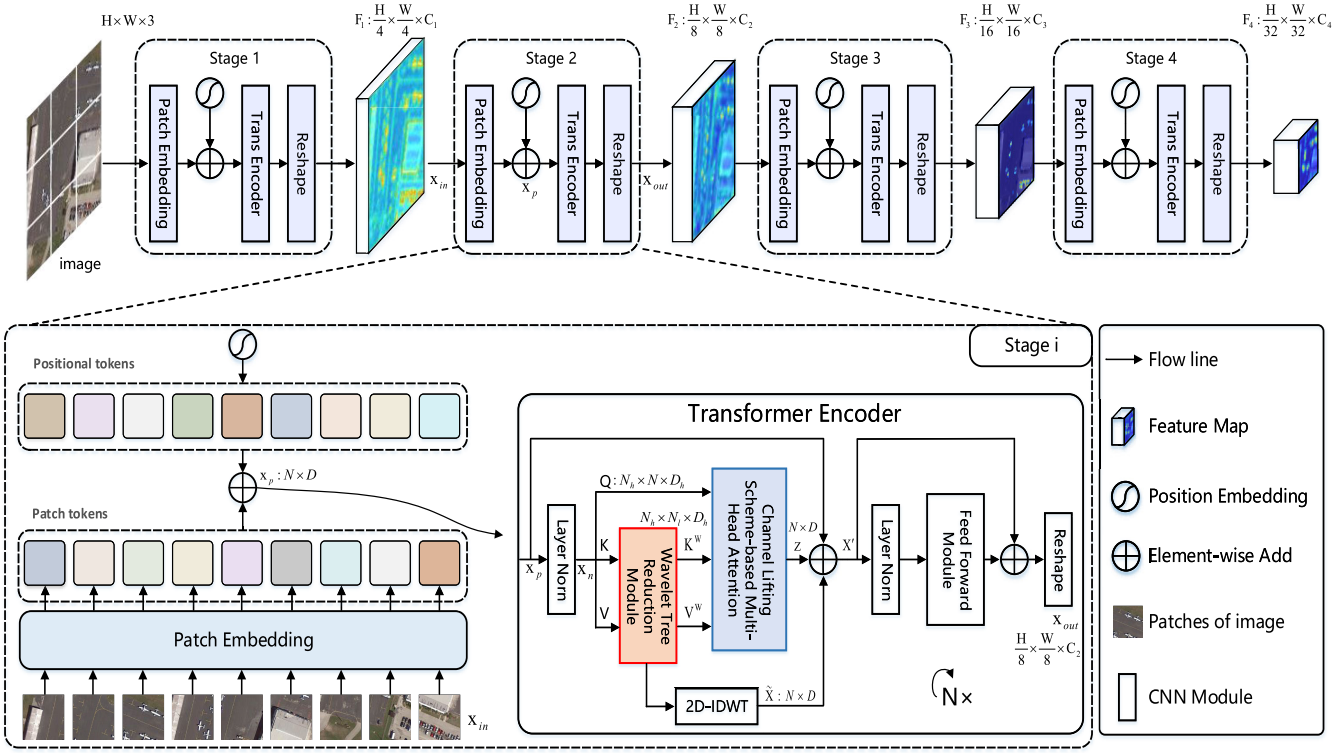

Fig. 4. Overall architecture of the proposed method. It is a Transformer [11] structure that can be applied in four stages of downstream tasks to effectively extract feature information and replace backbone. The transformer encoder contains our WTRM and CLSMHA modules as well as the wavelet inversion 2D-IDWT fusion module.

where M is the output of multihead attention and W^o is a linear transformation matrix. According to the analysis of the multihead attention calculation formula, its computational complexity is $O(H^2W^2D)$, which is the quadratic complexity of the input size. Obviously, this is very unfavorable for remote sensing visual tasks that take high-resolution images as inputs.

IV. Method

A. Overall Framework

The decoders of Transformer [17] are used to detect potential objects with similar areas and their positions on input features, which cannot be applied in object detection network alone. The encoders use the self-attention mechanism to conduct global modeling and extract features in another dimension by reconstructing heterogeneous features, which can meet the needs of the network. Besides, the simultaneous application of encoders and decoders will bring huge resource consumption. Therefore, our model is only based on the encoder structure of Transformer. As shown in Fig. 4, we construct a Transformer encoder structure based on wavelet tree structure and channel lifting scheme (CLS). The overall idea is to combine the multiresolution high- and low-frequency features of the wavelet tree and the long-short range combination formed by Transformer in the different heads of multihead attention. These features are aggregated into weighted heads to prioritize more crucial head features. Simultaneously, global context information modeling is performed from both frequency and spatial multidimensional perspectives, considering the distinct characteristics of each object for interaction.

Table II details architecture specifications for three variants of our WTHA-ViT with different model sizes, i.e., WTHA-ViT-S (small size), WTHA-ViT-M (medium size), and WTHA-ViT-L (large size). Multiple encoders are cascaded in four stages to replace the backbone of the object detection network.

Our WTHA-ViT comprises two main components: first, the WTRM combines the tree structure on wavelet frequencies with the long-range dependency attention in the Transformer, aiming to facilitate global contextual information modeling while efficiently representing multiscale and multifrequency information through a wavelet tree-based spatial reduction module. The second part is the CLSMHA. The CLS is applied to construct high- and low-frequency features distributed along the channel, and a weighted sum is performed on the multiple heads to prioritize the more important head features. This enables the different attention heads in the Transformer to perform distinct tasks and process features of varying scales.

The overall algorithm is shown in Algorithm 1. Denoting input and output of a certain stage by $\mathbf{x}_{\text{in}} \in \mathbb{R}^{(H\times W\times C)}$ and $\mathbf{x}_{\text{out}} \in \mathbb{R}^{((H/p)\times(W/p)\times D)}$. L, N, and c present the level of wavelet decomposition, the level of wavelet reconstruction, and the wavelet coefficient for each subband, respectively. The feature map $\mathbf{x}_{\text{in}}$ is fed into the patch embedding and position embedding layers, which is used to convert the original 2-D features into a sequence of 1-D vectors. $\mathbf{x}_p$ is then input into

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TABLE II

DETAILED ARCHITECTURE SPECIFICATIONS FOR THREE VARIANTS OF OUR WTHA-VIT WITH DIFFERENT MODEL SIZES, I.E., WTHA-VIT-S (SMALL SIZE), WTHA-VIT-M (MEDIUM SIZE), AND WTHA-VIT-L (LARGE SIZE). E_i, HEAD_i, C_i, P_i, AND TREE_i REPRESENT THE EXPANSION RATIO OF FEED-FORWARD LAYER, THE HEAD NUMBER, THE CHANNEL DIMENSION, THE SIZE OF EACH PATCH, AND THE LEVEL OF DECOMPOSITION OF THE WAVELET TREE IN EACH STAGE i, RESPECTIVELY

	Output Size	WTHA-ViT-S	WTHA-ViT-M	WTHA-ViT-L
Stage1	$\frac{H}{4} \times \frac{W}{4}$	$\begin{pmatrix} P_1=4 \\ E_1=8 \\ Tree_1=8 \\ Head_1=2 \\ C_1=64 \end{pmatrix} \times 3$	$\begin{pmatrix} P_1=4 \\ E_1=8 \\ Tree_1=8 \\ Head_1=2 \\ C_1=64 \end{pmatrix} \times 3$	$\begin{pmatrix} P_1=4 \\ E_1=8 \\ Tree_1=8 \\ Head_1=2 \\ C_1=64 \end{pmatrix} \times 3$
Stage2	$\frac{H}{8} \times \frac{W}{8}$	$\begin{pmatrix} P_2=2 \\ E_2=8 \\ Tree_2=8 \\ Head_2=4 \\ C_2=128 \end{pmatrix} \times 5$	$\begin{pmatrix} P_2=2 \\ E_2=8 \\ Tree_2=8 \\ Head_2=4 \\ C_2=128 \end{pmatrix} \times 5$	$\begin{pmatrix} P_2=2 \\ E_2=8 \\ Tree_2=8 \\ Head_2=4 \\ C_2=128 \end{pmatrix} \times 6$
Stage3	$\frac{H}{16} \times \frac{W}{16}$	$\begin{pmatrix} P_3=2 \\ E_3=4 \\ Tree_3=4 \\ Head_3=8 \\ C_1=256 \end{pmatrix} \times 6$	$\begin{pmatrix} P_3=2 \\ E_3=4 \\ Tree_3=4 \\ Head_3=10 \\ C_1=320 \end{pmatrix} \times 10$	$\begin{pmatrix} P_3=2 \\ E_3=4 \\ Tree_3=4 \\ Head_3=12 \\ C_1=384 \end{pmatrix} \times 14$
Stage4	$\frac{H}{32} \times \frac{W}{32}$	$\begin{pmatrix} P_4=2 \\ E_4=4 \\ Tree_4=1 \\ Head_4=14 \\ C_4=448 \end{pmatrix} \times 3$	$\begin{pmatrix} P_4=2 \\ E_4=4 \\ Tree_4=1 \\ Head_4=16 \\ C_4=512 \end{pmatrix} \times 3$	$\begin{pmatrix} P_4=2 \\ E_4=4 \\ Tree_4=1 \\ Head_4=16 \\ C_4=512 \end{pmatrix} \times 3$

Algorithm 1 Stage of WTHA-ViT

Input: feature map $\mathbf{x}_{in} \in \mathbb{R}^{(H\times W\times C)}$;
Output: feature map $\mathbf{x}_{out} \in \mathbb{R}^{(\frac{H}{\text{P}}\times\frac{W}{\text{P}}\times D)}$;
calculate embedding feature $\mathbf{x}_p$ by Eq.(13);
get LN feature $\mathbf{x}_n$ and Q, K, V feature by Eq.(14);
K, V input WTRM to get K^w, V^w by Eq.(1)-(10), (15);
L-lever DWT for K, V;
for i in range(L) **do**
  caculate K, V threshold T_0 by Eq.(21);
  if $A = \text{Max(Abs}(c)), c \in [LL_i, LH_i, HL_i, HH_i]$ **then**
    Tree_list$[i] \leftarrow A$
  end
  $T_0 = \frac{T_0}{2}$
end
for j in range(N) **do**
  $K^W, V^W +=$ IDWT(**Tree_list**$[L-j-1]$)
end
$Q \leftarrow Q$, $K^w \leftarrow K^W$, $V^w \leftarrow V^W$ by Eq.(16);
get $head_j$ in CLSMHA module by Eq.(17);
Perform a CLS on x_n to get Y by Eq.(22)-(27);
$HM(\cdot) \leftarrow Y$;
$\mathbf{z} \leftarrow HM(\cdot) \otimes head_j$;
$\tilde{\mathbf{x}} \leftarrow$ 2D-IDWT(K^w, V^w) by Eq.(19);
calculate $\mathbf{x}_{out}$ by Eq.(20);
return $\mathbf{x}_{out}$;

the encoder structure of the Transformer. It undergoes layer normalization and linear transformations to obtain the Q, K, and V features

$$\mathbf{x}_p = pa(\mathbf{x}_{\text{in}}, \xi) \oplus po(\mathbf{x}_{\text{in}}, \delta),\ \ \mathbf{x}_p \in \mathbb{R}^{(N\times D)} \tag{13}$$

$$\mathbf{x}_n = \text{LN}\big(\mathbf{x}_p, \theta\big),\ \ \mathbf{x}_n, Q, K, V \in \mathbb{R}^{(N\times D)}$$

$$Q, K, V = \mathbf{x}_n w^q, \mathbf{x}_n w^k, \mathbf{x}_n w^v \tag{14}$$

where $N = (HW/\text{P}^2)$, D is the hidden channel dimension of this stage. And po, po, and LN represent the patch embedding, position embedding, and layer normalization operation, respectively. Their trainable parameters are ξ, δ, θ and w^q, w^k, w^v are the transformation matrices of Q, K, and V. After performing a reshape operation on K and V, the WTS module uses (1)–(10) to carry out a L-level wavelet decomposition on them. The WTS organizes frequency subbands into a tree structure and selects the appropriate l-level ($l \in \{1, 2, \ldots, L\}$) downsampling frequency features K^W, V^W for each patch. Then, split Q, K, and V along the head

$$K^W, V^W = \text{WTS}(r(K, V), \varepsilon), K^W, V^W \in \mathbb{R}^{(N_l\times D)} \tag{15}$$

$$Q, K^w, V^w = f\big(Q, K^W, V^W\big)$$

$$K^w, V^w \in \mathbb{R}^{(N_h\times N_l\times D_h)}, Q \in \mathbb{R}^{(N_h\times N\times D_h)} \tag{16}$$

where r, f represent the reshape and flatten operation, respectively. ε is the trainable parameter of WTS module. $N_l = (h/2^l) \times (w/2^l)$, $h = (H/\text{P})$, and $w = (W/\text{P})$. In each head, self-attention is performed using scaled dot-product operation, and its result is denoted as head_j. The outputs from all heads are then concatenated along the channel dimension and transformed through a linear transformation to obtain the intermediate representation. Following (22)–(27), we apply the CLS to $\mathbf{x}_n$, resulting in Y. Next, we model the importance of each head and weight it to the output of the multihead attention, yielding the final output $\mathbf{z} \in \mathbb{R}^{(N\times D)}$

$$\text{head}_j = s\Big(Q_j K_j^{w\text{T}}/\sqrt{D_h}\Big)V_j^w, \text{head}_j \in \mathbb{R}^{(N\times D_h)} \tag{17}$$

$$\mathbf{z} = r\big(HM(Y, \zeta) \otimes c\big(\text{head}_1, \text{head}_2, \ldots, \text{head}_{N_h}\big)w^o\big) \tag{18}$$

where $\oplus$, $\otimes$, s, and c represent the element-wise add, the element-wise multiplication, Softmax, and Concat operation, respectively. HM is a head modeling operation and ζ is its

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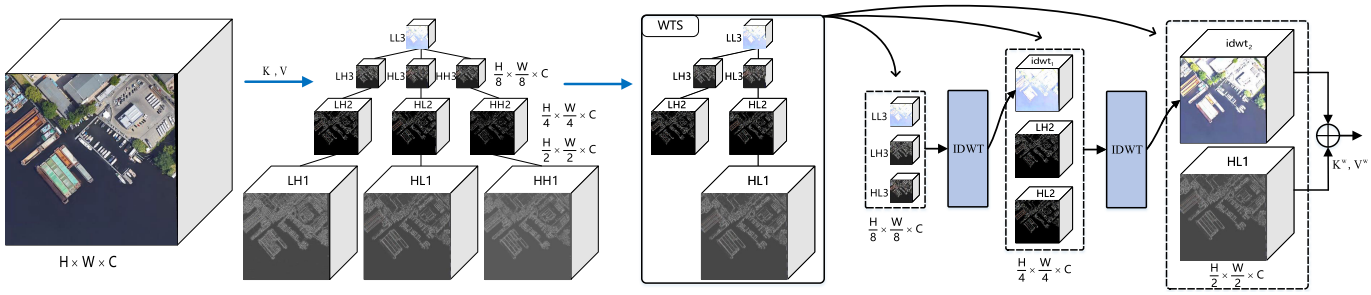

Fig. 5. WTRM. The input K, V features undergo multilevel wavelet decomposition, represented in a tree structure. Based on the WTS strategy, appropriate downsampled features K^W, V^W are selected from the wavelet decomposed features.

trainable parameter. $Q_j \in \mathbb{R}^{(N\times D_h)}$ and K_j^w, $V_j^w \in \mathbb{R}^{(N_l\times D_h)}$. Finally, according to (1), we perform a 2-D wavelet inverse transform on the downsampled K^w and V^w to reconstruct the features $\tilde{\mathbf{x}}$. Then, it is fused with the output $\mathbf{z}$ from the CLSMHA module and passed through a feed-forward neural network. Finally, after a reshape operation, it serves as the output of the stage

$$\tilde{\mathbf{x}} = \text{IDWT}(r(K^w, V^w)), \tilde{\mathbf{x}} \in \mathbb{R}^{(N\times D)} \tag{19}$$

$$\mathbf{x}_{\text{out}} = r\left(m\left(\mathbf{z} \oplus \tilde{\mathbf{x}}, \gamma\right)\right), \mathbf{x}_{\text{out}} \in \mathbb{R}^{(\frac{H}{P}\times\frac{W}{P}\times C)} \tag{20}$$

where m is the feed-forward neural network and γ is its trainable parameter.

B. Wavelet Tree Reduction Module

The WTRM shown in Fig. 5 takes the K and V features as input, and uses a spatial reduction module based on a wavelet tree structure for lossless downsampling learning. Here, we use wavelet decomposition and organize the multiple frequency features in the same direction into the same branch, thus forming a wavelet tree structure. Depending on the ratio $r = 2^l$ ($l \in \{1, 2, \ldots, L\}$) set by the network, the module outputs the downsampled features $K^W \in \mathbb{R}^{((N/r^2)\times D)}$ and $V^W \in \mathbb{R}^{((N/r^2)\times D)}$. As such, the overall computational cost of self-attention is significantly reduced by a factor r [i.e., $O((H^2W^2D/r^2))$].

For each patch, the K^W and V^W features extracted from it undergo a multilevel 2-D DWT. This transformation generates multiple resolution frequency subbands, which are organized into a tree structure. The root node of the tree corresponds to the LL frequency component obtained from the final level of decomposition. The selection of multiple subbands obtained through multilevel discrete wavelet decomposition is accomplished using the WTS strategy. This strategy considers multiple frequency feature information, enabling the removal of high-frequency noise while achieving the best approximation of the original feature.

This strategy bears similarity to the principle of embedded zero-tree wavelet (EZW) encoding [53]. Given a threshold T, the process begins at the root node of the wavelet tree, where the subband wavelet coefficient $\mathbf{x}_i$ is retained if $\mathbf{x}_i > T$, otherwise, it is discarded. The threshold T is initialized based on the formula of EZW coding, with the root node of the wavelet tree serving as the starting point

$$T_0 = 2^{\lfloor \log_2 \text{Max}|c_{i,j}| \rfloor} \tag{21}$$

where $c_{i,j}$ is the wavelet coefficient of each pixel corresponding to each subband, and Max() is the maximum operation. Starting from the lowest resolution frequency subband, the traversal continues until the resolution required by the network is reached. The threshold value $T_i = (T_{i-1}/2)$, $(i = 1, 2, 3, \ldots, L-1)$ is updated when different levels of nodes are traversed. If there are lower level subbands in the retained tree structure than the output level required by the network, these lower level subbands are first subjected to a 2-D discrete wavelet inverse transform to obtain subband features with the same resolution required by the network. Then, a feature fusion operation is performed with the retained subbands of the same resolution to obtain the downsampled K^W and V^W features. This method demonstrates the effectiveness in denoising and preserving essential features while reducing redundant information and enhancing interpretability.

C. CLS-Based Multihead Attention

Due to the correlation between channels in feature maps and inspired by FCANet [30], different channels in feature maps have varying sensitivities to different frequency components. To address this, we propose the CLS that constructs a low-frequency approximate component I_L and a high-frequency detailed component I_H distributed along the channel dimension, as shown in Fig. 6. The central concept of CLSMHA is that, since multihead attention also splits features along the channel dimension, each head corresponds to a portion of the original channel. After performing self-attention for each head, the results are concatenated to obtain the final attention output. However, the conventional multihead attention treats each head equally. According to the CLS, which associates high- and low-frequency features distributed along the channel dimension with each head of the multihead attention computation, we propose to weigh the importance of each head to focus on the features of more critical heads.

Not only do images have high- and low-frequency components, but the output feature maps of convolutional layers also contain high- and low-frequency components across channels. Due to the correlation between channels, lifting scheme [24], [36], [45], [54] can be applied along the channels to construct a

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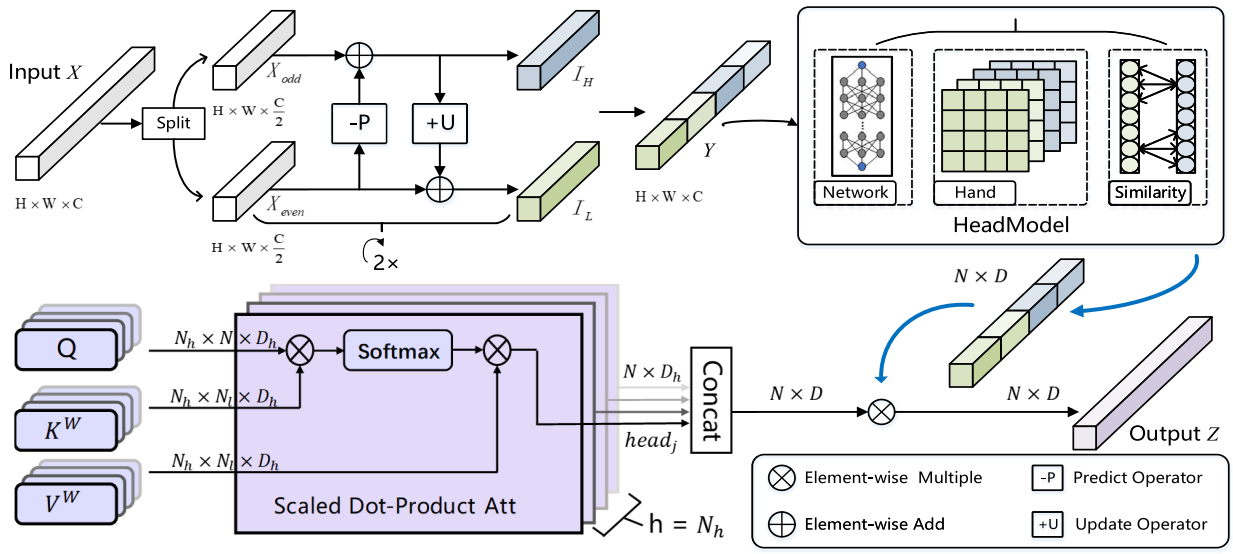

Fig. 6. CLSMHA. We use the CLS to obtain the high- and low-frequency features distributed along the channel, and then design three methods to model their head importance. The final result is weighted to the output of the individual head calculations in self-attention.

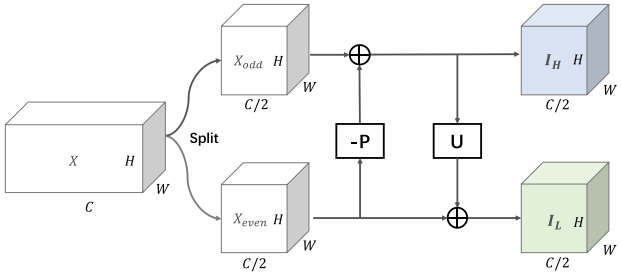

Fig. 7. CLS. The implementation process of the second-generation wavelet transform. It is quickly implemented in the spatial domain through split, predict, and update operations.

low-frequency approximate component I_L and high-frequency detailed component I_H distributed across the channels.

The CLS, illustrated in Fig. 7, processes the feature map $X \in \mathbb{R}^{(H\times W\times C)}$ as its input, where C, H, and W represent the channel, height, and width of the feature map, respectively. It generates the approximate subband c and the detailed subband d as outputs. The scheme comprises three steps: Split, Predict, and Update. In the Split step, X is split into two nonoverlapping parts. In this study, we perform even-odd sampling along the channel dimension to split the input X into two parts, resulting in the even index sample X_{even} and the odd index sample X_{odd}, which exhibit mutual correlation

$$X_{\text{odd}}, X_{\text{even}} = \text{Channel_Split}(X) \tag{22}$$

where $X_{\text{odd}}, X_{\text{even}} \in \mathbb{R}^{(H\times W\times (C/2))}$. As the odd and even index samples obtained from the split operation are correlated, given one set of these samples, it is possible to build a good predictor P for the other set by tracking the detailed subband d. Therefore, in the Predict step, X_{odd} can be predicted from X_{even}

$$I_H = d = X_{\text{odd}} - \text{P}(X_{\text{even}}) \tag{23}$$

$$\text{P} = \text{Tanh}(\text{conv}(\cdot)) \tag{24}$$

where P is the prediction operator. The final step is the Update operation. According to the principle, the approximate subband c should have the same average value as the input feature map. To achieve this, the approximate subband c is updated through an operator to refine the detailed subband d and optimize X_{even}

$$I_L = c = X_{\text{even}} + U(d) \tag{25}$$

$$U = \text{Tanh}(\text{conv}(\cdot)) \tag{26}$$

where U is the update operation and $I_L, I_H, c, d \in \mathbb{R}^{(H\times W\times (C/2))}$. Finally, I_L and I_H are concatenated along the channels using the Concat$(\cdot)$ operation, resulting in the output feature map $Y \in \mathbb{R}^{(H\times W\times C)}$

$$Y = \text{Concat}(I_L, I_H). \tag{27}$$

After obtaining high- and low-frequency features distributed along the channel dimension from CLS, it is crucial to model the importance of each frequency feature. We propose three methods for modeling the importance of each head: 1) Hand: manually designing weight parameters, where more attention weight are assigned to low-frequency features; 2) Network: learning weight parameters through a neural network, where a CNN is used to learn the channel importance of each head; and 3) Similarity: designing weight according to channel correlation, where the importance is modeled based on the correlation between the channels of each frequency feature.

The importance modeling results from the previous step are utilized to apply weights to the multiple heads. This process represents our head attention operation. Finally, according to (19), applying the inverse DWT (IDWT) to K^w and V^w obtains the reconstructed feature map $\tilde{x}$, which can preserve every detail in the input $\mathbf{x}_{\text{in}}$ while having negligible computational cost.

V. Experiments

We evaluate the effectiveness of our proposed framework on two remote sensing datasets: the optical DOTA-V1.0

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[37] dataset and the SAR image HRSID [38] dataset. In the ablation analysis, we conduct ablation experiments on the WTRM and CLSMHA modules and discuss the results in detail. In the quantitative experiments, we compare our WTHA-ViT with SOTA models and conduct sufficient ablation experiments to demonstrate the effectiveness of our proposed algorithm. Furthermore, we replace the backbone of multiple mainstream detectors with WTHA-ViT and compare it with the original network, further illustrating the stability of WTHA-ViT and its superior ability to extract feature information. In addition, we carry out feature visualization and model stability test. We evaluate the robustness of WTHA-ViT by adding noise to the test datasets, and the results and visual representation demonstrate the robustness of our framework. Finally, we select two large-scale area datasets for remote sensing object detection tasks, namely DIOR [59] and LEVIR [60], to explore the transferability and generalization ability of the model.

A. Dataset

1) Two Datasets to Verify the Performance of the Model: DOTA-V1.0 [37] is an aerial optical remote sensing image dataset designed for object detection. It comprises 2806 aerial images from different sensors, with a total of 188 282 object instances across 15 categories. The image sizes vary from 800 × 800 to 4000 × 4000 pixels. To facilitate processing, we crop the original images into smaller ones with a size of 1024 × 1024 and an overlap of 200 and 400 pixels. The training dataset consists of 38 233 images and the test dataset contains 12 951 images. This dataset offers ample data and diverse categories, making it suitable for reflecting practical scenarios.

HRSID [38] is a ship detection dataset specifically designed for high-resolution SAR images in remote sensing. The dataset includes 5604 high-resolution SAR ship images, with a total of 16 951 instances of ships. Each image has a fixed size of 800 × 800 pixels and belongs to a single category. The training dataset consists of 3642 images, while the test dataset contains 1962 images.

2) Large-Scale Datasets to Verify the Transferability: DIOR [59] dataset is a large-scale benchmark dataset for remote sensing image object detection, in which the image size is 800 × 800 pixels and the spatial resolution is 0.5–30 m. It contains 23 463 images and 190 288 object instances, covering 20 object categories, which is larger than the DOTA [37] dataset. The DIOR is divided into 11 725 images in the training verification set and 11 738 images in the test set.

LEVIR [60] dataset consists of a large number of high-resolution Google Earth images with 0.2–1.0 m pixels, and the size of each image is 800 × 600 pixels. The dataset consists of 21 952 images with three categories: plane, ship, and oiltank. There are 11 028 object instances in all images, including 4724 planes, 3025 ships, and 3279 oiltanks. The training set consists of 15 362 images and the test set contains 6590 images.

B. Experimental Setup

1) Implementation Details: We conduct experiments using our WTHA-ViT within the object detection network. The mmdetection [63] framework is employed with the Adamw optimizer as the optimization strategy, an initialization learning rate of $1e^{-4}$, and weight decay of $1e^{-4}$. We adjust the images in the HRSID dataset to (832, 832). The images in the DOTA-V1.0 dataset are kept at a size of (1024, 1024) for training and testing. We first load the pretraining weights trained on a large dataset ImageNet [64], initialize the backbone network, and conducted preliminary training in a large natural image object detection dataset COCO2017 [65]. Then, in the experimental, the pretraining weights preliminarily trained on COCO2017 [65] dataset are loaded for initialization and the newly added layers are initialized by Xavier initialization. Besides, the pretraining weight in all the comparative experiments is the same. For training, we utilize a 3× schedule (i.e., 36 epochs) in DOTA-V1.0 and the data augmentation operation of random flipping is used for the training set, with a flipping rate of 0.5. The network is trained with a batch size of 4 for DOTA-V1.0 and HRSID datasets on an Nvidia GeForce RTX 4090 GPU with 24 GB memory.

2) Evaluation Metrics: Some commonly used metrics are utilized in the experiments to analyze the effectiveness of the proposed model. Precision (P): the probability of the actual positive sample among all predicted positive samples. Precision helps to highlight the relevance of results. Recall (R): the likelihood of being anticipated to be positive among actual positive examples. The calculating formulas are

$$\mathrm{P} = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FP}}, \quad \mathrm{R} = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FN}} \tag{28}$$

where TP, FP, and FN denote the numbers of true positives, false positives, and false negatives, respectively.

The average precision (AP) metric considers precision and recall. The value of AP is the area under the precision–recall (P–R) curve for each category and the calculation formula is as follows:

$$\mathrm{AP} = \int_0^1 \mathrm{P}(\mathrm{R}) d\,R. \tag{29}$$

For coco format dataset, 100 points are sampled on the horizontal axis of P–R curve for calculation. The threshold of IoU increases from 0.5 to 0.95, and the AP value is calculated every 0.05, and the average value of all results is taken as the final result, which is the AP metric used in our experiment.

To evaluate the detection accuracy of the proposed model, we use the mean average precision (mAP) of object detection as the evaluation metric. For object detection, the larger the AP and mAP, the higher the accuracy of the detection model. The calculation formula for mAP is as follows:

$$\mathrm{mAP} = \frac{1}{n} \sum_{i=1}^{n} \mathrm{AP}_i \tag{30}$$

where i denotes the ith category, and n denotes the total number of categories. AP_i is the AP of the ith category. In this article, our AP metric also refers to mAP when the dataset contains multiple categories. AP_s, AP_m, and AP_l, respectively, represent the AP of small, medium, and large-scale objects. Among them, the pixel range of the small-scale object is [0, 32 × 32), the medium-scale object is [32 × 32, 96 × 96),

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TABLE III

PERFORMANCES (%) OF VARIOUS VISION BACKBONES ON DOTA-V1.0 DATASET FOR THE DOWNSTREAM TASKS OF OBJECT DETECTION. FOR OBJECT DETECTION TASK, WE EMPLOY RETINANET AS THE OBJECT DETECTOR, AND THE AP AT DIFFERENT IOU THRESHOLDS AND THREE DIFFERENT OBJECT SIZES [I.E., SMALL, MEDIUM, AND LARGE (S/M/L)] ARE REPORTED FOR EVALUATION

Backbone	Params	GFLOPs	Precision	Recall	F_1	AP	AP_{50}	AP_{75}	AP_s	AP_m	AP_l
ResNet50 [55]	36.39M	215.46G	69.36	81.67	0.7501	45.8	70.2	50.2	19.8	50.3	55.2
PVT-T [22]	21.62M	158.11G	72.35	83.70	0.7761	47.3	73.4	52.6	19.5	50.5	61.0
Swin-T [13]	37.11M	221.64G	73.46	83.63	0.7822	49.8	74.9	56.4	21.2	54.7	61.2
PVT-S [22]	32.87M	189.22G	72.63	83.47	0.7767	47.5	73.6	52.0	18.7	49.9	61.2
Twins-SVT-S [21]	32.44M	185.68G	72.37	82.38	0.7705	47.6	73.0	52.6	18.0	50.9	61.1
BoTNet [44]	35.66M	212.91G	73.53	83.22	0.7808	51.2	74.2	55.6	20.5	55.1	61.8
UniFormer-S [56]	29.93M	199.68G	73.67	83.67	0.7835	50.0	74.3	56.1	21.1	55.3	61.9
DAT++-T [57]	32.34M	211.67G	74.13	82.91	0.7827	51.3	74.3	57.1	16.0	55.8	**69.4**
LIT-T [58]	28.27M	196.00G	73.32	83.37	0.7802	49.0	74.1	54.7	19.0	54.1	62.2
WaveViT-S [23]	28.11M	216.51G	74.44	82.64	0.7833	50.3	74.5	56.2	21.1	53.0	64.9
SpectFormer-S [26]	50.22M	206.92G	73.49	81.22	0.7716	50.7	73.4	56.8	20.8	54.1	65.7
GFNet-T [27]	54.29M	178.82G	71.98	82.06	0.7671	48.3	72.4	54.5	17.9	54.2	61.0
AFNO-T [28]	23.51M	178.82G	73.88	82.71	0.7805	50.6	74.6	56.6	19.5	56.0	63.3
WTHA-ViT-S(**ours**)	35.07M	228.46G	**75.15**	**84.62**	**0.7962**	**52.4**	**76.5**	**60.5**	**23.1**	**58.1**	64.4
ResNet101 [55]	55.39M	293.36G	72.97	82.74	0.7755	47.8	73.3	53.5	19.7	53.4	58.2
PVT-M [22]	52.59M	234.67G	74.72	84.11	0.7914	50.4	75.5	56.7	20.9	54.0	64.6
Swin-S [13]	58.43M	314.38G	73.38	84.07	0.7828	47.1	74.4	51.6	18.5	50.7	61.5
Twins-SVT-B [21]	64.91M	304.43G	74.31	84.12	0.7891	50.7	75.0	57.2	20.6	56.2	63.7
UniFormer-B [56]	58.17M	293.07G	75.09	84.29	0.7942	52.7	76.2	59.7	22.6	59.2	63.0
DAT++-S [57]	61.95M	317.87G	75.55	83.18	0.7918	52.3	75.1	58.3	17.6	56.6	**69.7**
LIT-S [58]	57.56M	314.09G	75.09	83.33	0.7900	50.7	74.7	56.1	15.0	55.3	69.5
WaveViT-B [23]	38.24M	266.59G	74.41	83.21	0.7856	50.3	74.6	55.7	18.0	54.6	66.5
SpectFormer-B [26]	61.08M	256.99G	74.68	83.47	0.7883	51.8	75.2	57.1	20.1	57.0	67.4
GFNet-S [27]	89.10M	284.02G	75.11	83.59	0.7912	52.8	76.1	59.7	22.5	58.1	64.5
AFNO-S [28]	43.48M	284.02G	74.76	83.16	0.7874	51.8	75.3	58.2	20.8	56.4	65.2
WTHA-ViT-M(**ours**)	53.69M	286.12G	**76.73**	**84.42**	**0.8039**	**55.0**	**77.6**	**62.8**	**24.8**	**59.5**	68.9
ResNext101 [55]	94.12M	454.36G	73.51	83.05	0.7799	49.7	74.2	55.0	19.6	54.4	63.3
PVTv2-b5 [22]	90.34M	323.70G	75.56	83.66	0.7940	52.3	77.1	58.6	19.6	55.6	68.7
Swin-B [13]	97.04M	461.06G	75.24	84.12	0.7943	49.6	76.2	54.9	19.9	53.7	65.0
Twins-SVT-L [21]	108.56M	443.32G	75.20	84.10	0.7940	51.3	75.8	57.5	18.0	57.4	66.0
UniFormer-L [56]	108.23M	535.37G	75.41	83.55	0.7927	54.0	77.5	62.7	20.0	60.9	68.3
DAT++-B [57]	102.96M	465.79G	75.58	83.56	0.7937	53.4	75.9	60.2	20.4	59.7	70.2
WaveViT-L [23]	62.33M	417.16G	75.44	84.23	0.7959	53.6	76.4	59.3	19.8	58.9	69.2
SpectFormer-L [26]	98.88M	390.50G	75.39	84.02	0.7947	52.9	76.0	60.7	19.9	59.7	68.8
GFNet-B [27]	137.05M	367.33G	75.66	83.33	0.7931	53.8	76.3	61.1	**22.9**	60.7	65.1
AFNO-B [28]	65.14M	367.33G	76.30	84.16	0.8004	54.0	76.6	62.3	22.0	59.2	68.2
Swin-L [13]	206.70M	879.46G	75.86	83.75	0.7961	53.4	77.4	61.0	20.0	58.2	69.6
WTHA-ViT-L(**ours**)	77.68M	353.06G	**77.39**	**84.76**	**0.8091**	**55.6**	**78.6**	**63.1**	21.1	**61.2**	**71.3**

We group the vision backbones into three categories, and all backbones within each category shares similar GFLOPs: Small (GFLOPs$<$230), Base (230$\leq$GFLOPs$<$320), Large (320$\leq$GFLOPs).

and the large-scale object is [96 $\times$ 96, $+\infty$). $AP_{50:95}$, AP_{50}, and AP_{75} represent the AP when the threshold range of IoU varies from 0.5 to 0.95, the threshold is fixed at 0.5, and the threshold is fixed at 0.75. The AP metric in this article refers to $AP_{50:95}$.

We add $F1$-scores to weigh P and R to further measure the quality of the model. It is calculated by the following formula:

$$F_1 = \frac{2\mathrm{PR}}{\mathrm{P}+\mathrm{R}} \in [0, 1]. \tag{31}$$

In addition, some metrics that appear in the article are as follows. Floating point operations (FLOPs) refers to the total number of floating-point operations performed during a calculation. It is used to measure the complexity of the computation task and the amount of computation. Floating-point operations refer to mathematical operations involving floating-point numbers, such as addition, subtraction, multiplication, and division. Params refers to the total number of parameters that need to be trained in the network model.

C. Comparison With Similar Architectures

In this section, we evaluate the performance of WTHA-ViT on two datasets. The mainstream detector RetinaNet [67] was used for object detection task, and we replaced the backbone in RetinaNet [67] with WTHA-ViT for evaluation.

The results on DOTA-V1.0 dataset are shown in Table III, which indicates that WTHA-ViT outperforms current state-of-the-art transformer-based algorithms in multiple metrics. WTHA-ViT-L model's AP_{50} values are 78.6%, reaching the SOTA level. Besides, we use ATSS [61] and Cascade RCNN [62] detectors for the DOTA-V1.0 and HRSID dataset, as shown in Table IV. In HRSID dataset, due to the small number of large objects and sample imbalance compared with other object sizes, detection biases may occur and the advantages of our algorithm cannot be reflected in this metric. And all ours and other advanced models perform poorly on AP_l metric, which is caused by datasets and has nothing to do with models.

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TABLE IV

PERFORMANCES OF VARIOUS VISION BACKBONES ON DOTA-V1.0 AND HRSID DATASET FOR OBJECT DETECTION. TWO KINDS OF OBJECT DETECTORS, I.E., ATSS [61] AND CASCADE RCNN [62] IN MMDETECTION [63] ARE ADOPTED FOR EVALUATION. WE REPORT THE BOUNDING BOX AP IN DIFFERENT IOU THRESHOLDS

Backbone	Method	Params	GFLOPs	DOTA-V1.0						HRSID					
				P	R	F_1	AP	AP_{50}	AP_{75}	P	R	F_1	AP	AP_{50}	AP_{75}
ResNet50 [55]	ATSS [61]	31.92M	206.86G	71.91	83.62	0.7732	48.1	72.3	53.4	80.29	89.80	0.8478	58.5	84.7	65.0
PVT-T [22]		20.68M	150.41G	74.34	84.75	0.7920	49.0	74.7	55.1	81.90	92.96	0.8708	62.2	87.0	70.0
Swin-T [13]		35.58M	213.79G	75.01	84.55	0.7949	49.2	75.7	55.1	82.32	92.86	0.8727	62.2	87.6	70.5
Twins-SVT-S [21]		31.51M	177.98G	74.19	83.99	0.7879	49.0	75.0	55.1	79.66	89.06	0.8410	57.8	83.8	64.2
BoTNet [44]		31.19M	204.30G	76.12	84.23	0.7997	51.9	76.5	59.7	82.44	92.13	0.8702	63.9	87.8	70.7
UniFormer-S [56]		29.00M	191.98G	75.84	85.20	0.8025	52.1	76.2	59.0	82.60	92.50	0.8727	64.7	88.0	72.6
DAT++-T [57]		68.40M	233.02G	74.62	80.49	0.7744	51.6	75.7	58.7	81.97	91.61	0.8652	62.3	87.1	69.5
LIT-T [58]		27.34M	188.30G	74.16	83.95	0.7875	49.7	74.1	55.9	79.87	89.06	0.8422	56.3	84.1	62.3
WaveViT-S [23]		27.33M	208.85G	76.51	84.47	0.8029	52.2	76.9	58.9	81.88	91.05	0.8622	62.8	87.0	69.9
SpectFormer-S [26]		49.43M	199.25G	76.50	84.40	0.8026	51.4	77.1	58.4	82.53	92.79	0.8736	61.8	87.9	69.9
GFNet-T [27]		53.35M	171.12G	75.45	83.47	0.7926	52.6	76.2	60.5	81.69	91.61	0.8637	62.6	86.7	70.7
AFNO-T [28]		22.57M	171.12G	75.76	84.19	0.7975	53.3	76.0	60.6	82.04	92.11	0.8678	62.9	87.2	70.9
WTHA-ViT-S(**ours**)		34.28M	220.80G	**77.12**	**85.56**	**0.8112**	**54.1**	**78.5**	**61.6**	**83.45**	**94.09**	**0.8845**	**66.8**	**90.7**	**75.6**
ResNet50 [55]	Cascade RCNN [62]	68.97M	239.13G	71.59	80.75	0.7589	47.0	73.3	52.0	78.19	84.45	0.8120	61.0	81.7	69.5
PVT-T [22]		57.68M	179.46G	73.13	82.66	0.7760	47.5	75.8	51.6	80.52	86.81	0.8355	63.8	84.2	73.5
Swin-T [13]		72.59M	243.37G	73.90	83.35	0.7834	49.0	77.2	54.8	81.45	87.08	0.8417	65.8	85.5	75.2
Twins-SVT-S [21]		68.51M	207.03G	75.50	81.66	0.7846	52.3	77.0	59.0	81.96	87.76	0.8476	62.8	84.8	71.9
BoTNet [44]		69.44M	236.57G	76.25	81.96	0.7900	53.2	77.5	60.4	81.88	87.27	0.8449	63.9	85.3	73.4
UniFormer-S [56]		66.00M	221.03G	75.94	82.28	0.7900	53.6	77.2	61.8	82.24	87.05	0.8458	65.2	85.2	74.8
DAT++-T [57]		68.40M	233.02G	78.33	80.40	0.7935	**57.2**	78.0	**65.3**	81.61	86.04	0.8377	63.6	84.3	72.7
LIT-T [58]		64.34M	217.35G	76.57	83.09	0.7970	53.1	78.1	60.5	81.05	85.22	0.8308	63.2	83.5	73.1
WaveViT-S [23]		64.33M	237.90G	77.33	81.06	0.7915	55.2	77.6	63.1	82.33	86.27	0.8425	66.5	84.9	76.2
SpectFormer-S [26]		86.43M	228.30G	77.17	81.45	0.7925	53.7	77.8	61.3	82.89	86.29	0.8456	65.5	84.7	75.3
GFNet-T [27]		90.35M	200.17G	77.00	81.86	0.7936	55.0	78.5	64.4	81.21	84.18	0.8267	63.0	82.3	72.0
AFNO-T [28]		59.57M	200.17G	77.42	81.68	0.7949	57.0	77.9	65.2	81.91	85.03	0.8344	63.5	83.3	73.1
WTHA-ViT-S(**ours**)		71.28M	249.85G	**78.64**	**83.60**	**0.8104**	56.9	**80.2**	65.1	**83.32**	**88.20**	**0.8569**	**69.2**	**86.5**	**79.5**

Fig. 8 is the visualization of the detection results of the HRSID dataset, from which it is clear that our method has the fewest missing ships and false alarms. In addition, we analyzed the loss curves and convergence speed of different backbone networks in the RetinaNet [67] detector on DOTA-V1.0 dataset and in the ATSS [61] detector on HRSID dataset, as shown in Fig. 9. It can be intuitively seen that our WTHA-ViT has some advantages compared with similar ViT methods, and its convergence speed is faster than most methods. Specifically, in DOTA-V1.0 dataset, our WTHA-ViT converges much faster than LIT [58], GFNet [27] and SpectFormer [26], which are similar to ViT. At the same time, in the HRSID dataset, our WTHA-ViT converges significantly faster than the five VIT-like models: Twins [21], LIT [58], DAT++ [57], AFNO [28], and WaveViT [23]. Fig. 10 shows their P–R curves, and the curves of our method are better than other comparison methods.

We compare the performance of three versions of WTHA-ViT (S, M, L) with several state-of-the-art algorithms using the Retinanet [67] detector in DOTA-V1.0 dataset. The evaluation metrics GFLOPs and parameters are calculated by inputting images with sizes of (1024, 1024, 3) into the network. As shown in Fig. 11(a) and (b), our model can achieve performance improvements while balancing the trade-off between performance and computational complexity compared to other advanced models. It can be seen from the figure that, for instance, when considering the FLOPs metric, the FLOPs required by WTHA-ViT-S is 228.46G, whereas PVT-M demands 234.67G FLOPs. Despite the lower FLOPs, our WTHA-ViT-S achieves an AP_{50} value of 76.5%, surpassing PVT-M's 75.5%. At the same time, in the Params metric, WTHA-ViT-S uses only 35.07M Params, which is significantly smaller than PVT-M's 53.69M Params. Although our model may have lost some computational cost, it is still able to focus on the head regions of interest and thereby reduce feature redundancy. Hence, we relatively reduce the channel dimension of the output feature maps at each stage of the network compared to other advanced models (e.g., Swin-L [13] has 1536 output channels in stage 4, and Twins-SVT-L [21] has 1024 output channels in stage 4) when designing the WTHA-ViT model.

For the task of oriented object detection (OOD) in remote sensing scenes, unlike previous comparative experiments, it employs annotated bounding boxes that incorporate rotation, which better aligns with remote sensing object. As shown in Table V, we use our WTHA-ViT to replace the backbone in our model specifically for remote sensing scenarios, compare it with the advanced one-stage and two-stage OOD methods on the DOTA-V1.0 dataset, and report detection results for each category as well as mAP metrics. Our model outperforms

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TABLE V

COMPARISONS WITH ADVANCED ONE-STAGE AND TWO-STAGE OOD METHODS ON THE DOTA DATASET. THE DETECTION RESULTS OF EACH CATEGORY ARE REPORTED TO BETTER DEMONSTRATE WHERE THE PERFORMANCE GAINS COME FROM. ◇ REPRESENTS THE TRANSFORMER-BASED MODEL. OUR WTHA-VIT-S IS USED IN THE ORIENTED RCNN [66] NETWORK, REPLACING THE ORIGINAL BACKBONE NETWORK WITH OUR WTHA-VIT-S

Model	Backbone	PL	BD	BR	GTF	SV	LV	SH	TC	BC	ST	SBF	RA	HA	SP	HC	mAP
One-stage																	
RetinaNet-O [67]	ResNet50	88.67	77.62	41.81	58.17	74.58	71.64	79.11	90.29	82.18	74.32	54.75	60.60	62.57	69.57	60.64	68.43
DRN [68]	H-104	88.91	80.22	43.52	63.35	73.48	70.69	84.94	90.14	83.85	84.11	50.12	58.41	67.62	68.60	52.50	70.70
R^3Det [69]	ResNet152	89.49	81.17	50.53	66.10	70.92	78.66	78.21	90.81	85.26	84.23	61.81	63.77	68.16	69.83	67.17	73.74
RSDet	ResNet152	**90.10**	82.00	53.80	68.50	72.20	78.70	73.60	**91.20**	87.10	84.70	64.30	68.20	66.10	69.30	63.70	74.10
S^2-Net [70]	ResNet50	89.11	82.84	48.37	71.11	78.11	78.39	87.25	90.83	84.90	85.64	60.36	62.60	65.26	69.13	57.94	74.12
Oriented RepPoints [71]	ResNet50	87.02	83.17	54.13	71.16	80.18	78.40	87.28	90.90	85.97	86.25	59.90	70.49	73.53	72.27	58.97	75.97
ADT-Det [72]	ResNet152	89.71	84.71	**59.63**	80.94	**80.30**	83.53	**88.94**	90.86	87.06	**87.81**	70.72	70.92	78.66	79.40	65.99	79.95
KLD [73]	ResNet152	89.92	**85.13**	59.19	81.33	78.82	84.38	87.50	89.80	**87.33**	87.00	**72.57**	71.35	77.12	79.34	**78.68**	80.63
S^2-Net+FRIoU [74]	ResNet50	89.83	85.08	56.17	**83.11**	79.95	**84.41**	88.78	90.78	85.66	87.70	72.05	**73.08**	**81.54**	**81.64**	71.23	**80.74**
Two-stage																	
Faster-RCNN-O [75]	ResNet50	88.44	73.06	44.86	59.09	73.25	71.49	77.11	90.84	78.94	83.90	48.59	62.95	62.18	64.91	56.18	69.05
SCRDet [76]	ResNet101	89.98	80.65	52.09	68.36	68.36	60.32	72.41	90.85	87.94	86.86	65.02	66.68	66.25	68.24	65.21	72.61
ARS-DETR◇ [77]	ResNet50	86.61	77.26	48.84	66.76	78.38	78.96	87.40	90.61	82.76	82.19	54.02	62.61	72.64	72.80	64.96	73.79
RoI Trans. [78]	ResNet101	88.65	82.60	52.53	70.87	77.93	76.67	86.87	90.71	83.83	82.51	53.95	67.61	74.67	68.75	61.03	74.61
Gliding Vertex [79]	ResNet101	89.64	85.00	52.26	77.34	73.01	73.14	86.82	90.74	79.02	86.81	59.55	70.91	72.94	70.86	57.32	75.02
DODet [80]	ResNet50	89.34	84.31	51.39	71.04	79.04	82.86	88.15	**90.90**	86.88	84.91	62.69	67.63	75.47	72.22	45.54	75.49
RODFormer◇ [81]	ViT-Based	89.76	79.64	56.61	71.57	78.60	85.29	**89.93**	90.53	87.73	83.05	60.19	60.34	66.03	69.75	64.95	75.60
Oriented RCNN [66]	ResNet50	89.46	82.12	54.78	70.86	78.93	83.00	88.20	**90.90**	87.50	84.68	63.97	67.69	74.94	68.84	52.28	75.87
AQS◇ [82]	Swin-Tiny	89.15	82.42	55.30	73.29	77.90	80.41	87.76	90.87	84.23	85.38	60.14	64.31	72.71	73.74	65.29	76.23
ReDet [83]	ReResNet50	88.79	82.64	53.97	74.00	78.13	84.06	88.04	90.89	87.78	85.75	61.76	60.39	75.96	68.07	63.59	76.25
CLT-Det◇ [3]	ResNet101	89.31	85.69	53.97	77.11	79.66	79.01	88.55	90.89	85.36	86.56	63.92	68.47	75.65	70.65	66.91	77.45
AO2-DETR◇ [84]	ResNet50	89.95	84.52	56.90	74.83	80.86	83.47	88.47	90.87	86.12	**88.55**	63.21	65.09	79.09	82.88	73.46	79.22
DEA [85]	ReResNet50	89.92	83.84	59.65	79.88	80.11	**87.96**	88.17	90.31	88.93	88.46	68.93	65.96	78.04	79.69	75.78	80.37
FDOL [86]	ReResNet50	88.90	84.57	60.73	80.83	78.42	85.82	88.33	**90.90**	88.28	86.93	71.44	67.13	79.00	80.35	74.59	80.41
AOPG [87]	ResNet50	89.88	85.57	60.90	81.51	78.70	85.29	88.85	90.89	87.60	87.65	71.66	68.69	82.31	77.32	73.10	80.66
AOPG+FRIoU [74]	ResNet50	90.13	85.85	60.04	80.90	79.21	85.34	88.73	90.88	86.35	87.55	70.22	72.09	83.40	82.12	72.55	81.02
RVSA◇ [88]	ViTAE-B	89.40	83.94	59.76	82.10	**81.73**	85.32	88.88	90.86	85.69	87.65	63.70	69.94	84.72	**84.16**	79.90	81.18
ours◇	WTHA-ViT-S	**90.50**	**87.60**	**65.60**	**83.70**	73.40	87.70	89.70	**90.90**	**89.00**	62.80	**85.50**	**77.50**	**85.20**	66.50	**81.80**	**81.20**

PL: plane. BD: baseball diamond. BR: bridge. GTF: ground track field. SV: small vehicle. LV: large vehicle. SH: ship. TC: tennis court. BC: baseball court. ST: storage tank. SBF: soccer ball filed. RA: roundabout. HA: harbor. SP: swimming pool. HC: helicopter.

other state-of-the-art remote sensing methods in OOD task.

We utilize several curves, namely C75, C50, the localization error curve (Loc), supercategory false positives (Sim), and the confusion with background curve (BG), as illustrated in Fig. 12. The C75 curve represents the precision–recall curve (PRC) at an IoU of 0.75, with the area under the curve corresponding to the AP_{75} metric. Similarly, the C50 curve is also defined in this way. The Loc curve refers to the PRC at an IoU of 0.1, where localization error results are disregarded, except for duplicate detections. The Loc curve is drawn to assess the impact of localization error on the detectors. Since the supercategory and subcategory are the same, the Sim curve coincides with the Loc curve. The Oth curve represents the PRC after eliminating all confusing categories. After removing all false alarm examples, a new PRC called BG is plotted. FN denotes the PRC after removing all remaining errors (resulting in AP $=$ 1). The green area represents the performance degradation caused by localization errors. The yellow area indicates the AP increment achieved by removing false alarms. At last, the red area signifies the AP improvement obtained by adding missing detections.

As shown in Fig. 12(a) and (b), RVSA [88] and our WTHA-ViT in Table V, it can be seen that the areas surrounded by the curves of C75 and C50 of our algorithm are larger than RVSA [88]. The Loc (green area), BG (yellow area), and FN (red area) curves can also directly reflect the performance of the model. The smaller the area of these three colors, the higher the performance of the corresponding model and our model is better than RVSA among these three curves, especially the FN curve. In addition, Fig. 12(c) and (d) are the inference results and RVSA [88] missed many vehicles. In contrast, the inference results of our WTHA-ViT on land are much better than RVSA [88]. We selected the PRC at IoU $=$ 0.5 (C50) to calculate the AP_{50} and $F1$-scores metrics. It is visually evident from Fig. 12 that our WTHA-ViT has a higher AP_{50} value than RVSA [88], indicating that the $F1$-scores of our WTHA-ViT is correspondingly greater than RVSA [88]. From the above results, we can draw a conclusion that our network has performance superiority in remote sensing object detection.

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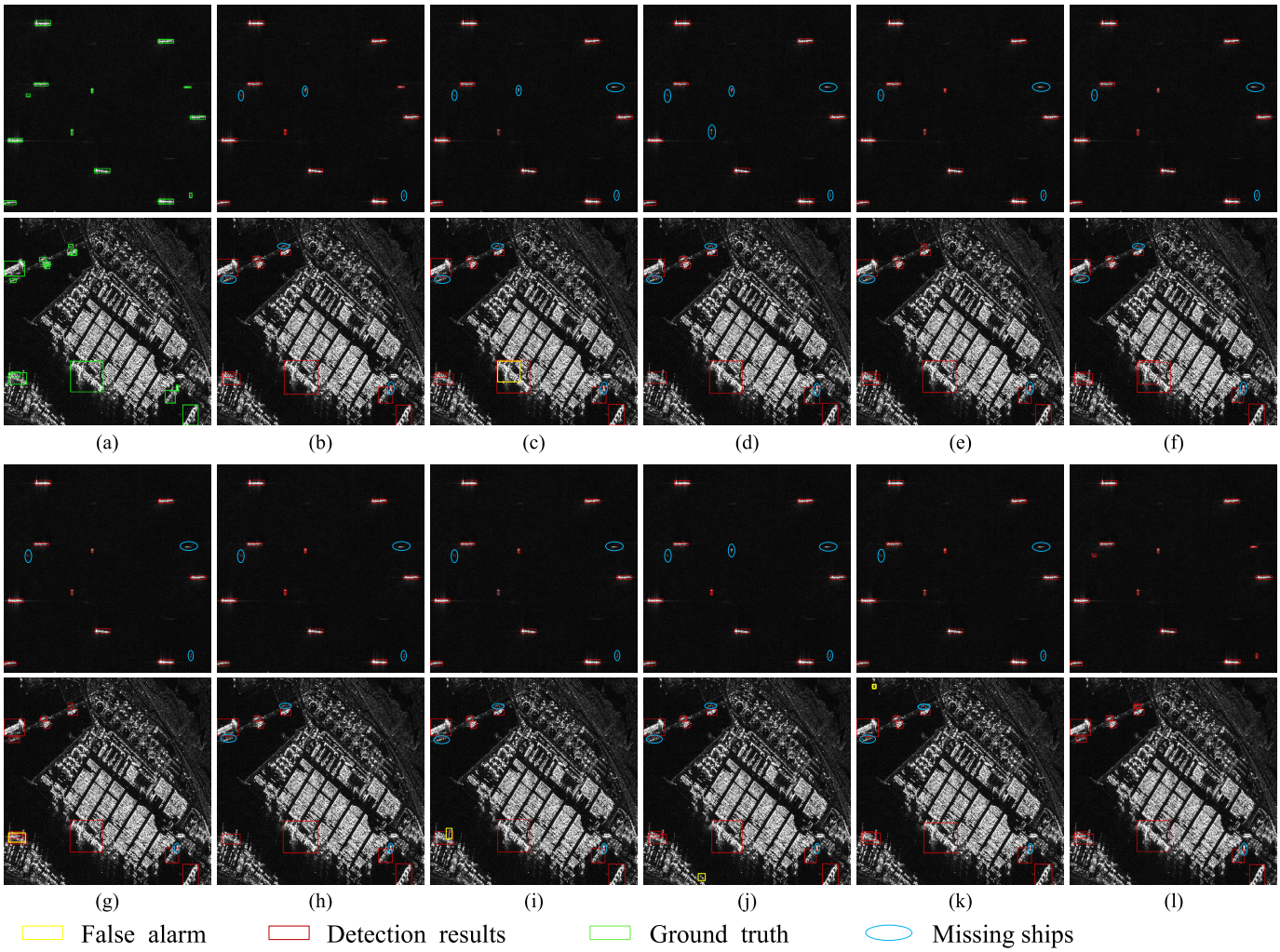

Fig. 8. Visualization of the inference results for different models on HRSID dataset. (a) Original images with GT. (b) PVT-T [22]. (c) Swin-T [13]. (d) Twins-SVT-S [21]. (e) Wavevit-S [23]. (f) LIT-T [58]. (g) DAT++-T [57]. (h) UniFormer-S [56]. (i) GFNet-T [27]. (j) AFNO-T [28]. (k) SpectFormer-S [26]. (l) WTHA-ViT-S (ours).

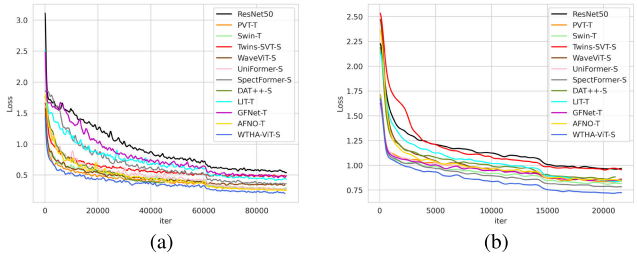

Fig. 9. Comparison of loss function on DOTA-V1.0 and HRSID datasets of different backbone. (a) Loss in DOTA-V1.0. (b) Loss in HRSID.

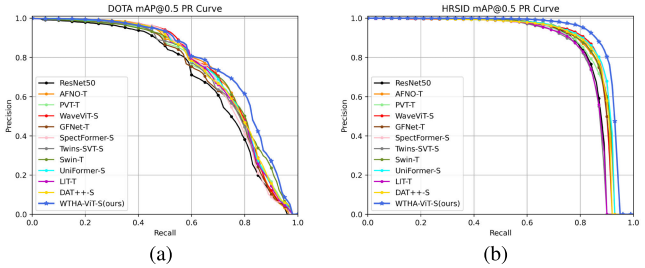

Fig. 10. P–R curves of different backbone models. (a) Results of RetinaNet [67] detector in DOTA-V1.0 dataset. (b) Results of ATSS [61] detector in HRSID dataset.

D. Implementation on Mainstream Detectors

To further demonstrate the stability and versatility of WTHA-ViT, and its ability to better extract feature information, we apply our WTHA-ViT to multiple mainstream detectors by replacing their backbone and compare the results with the original networks. The quantitative results on various detectors are shown in Table VI.

According to the Table VI, our proposed framework can be effectively adapted to the backbone networks of various object detectors, including single-stage networks such as FCOS [89], two-stage networks such as Cascade RCNN [62], Dynamic RCNN [93], Mask RCNN [94], Sparse RCNN [95], and Libra RCNN [96] as well as anchor-based networks like ATSS [61], AutoAssign [90], GFL [91], and Guided Anchor [92] networks. Moreover, we observe that the detection performance of these detectors is improved to some extent when using the WTHA-ViT-S. Our experiments on DOTA-V1.0 and HRSID datasets have demonstrated that our algorithm can perfectly replace the feature extraction module. In addition, as shown

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TABLE VI

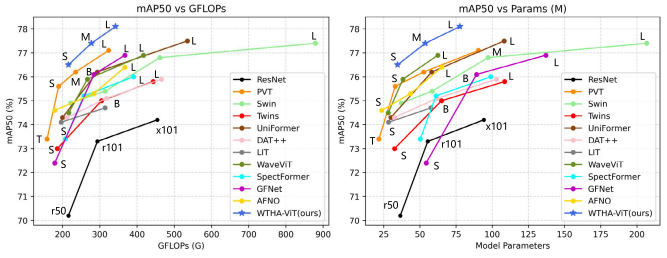
QUANTITATIVE EXPERIMENTAL RESULTS (%) OF WTHA-VIT-S ON VARIOUS DETECTORS. † REPRESENTS THE EXPERIMENT RESULTS (%) IN WHICH THE BACKBONE OF THE ORIGINAL NETWORK BECOMES WTHA-VIT-S

Method	Params	GFLOPs	DOTA-V1.0						HRSID					
			P	R	F_1	AP	AP_{50}	AP_{75}	P	R	F_1	AP	AP_{50}	AP_{75}
FCOS [89]	31.87M	201.99G	73.46	84.30	0.7851	47.9	75.2	54.4	80.78	90.97	0.8557	55.3	85.4	60.8
FCOS†	34.28M	220.80G	75.47	85.49	0.8017	52.9	77.6(+2.4)	59.7	83.86	93.58	0.8835	65.8	89.8(+4.4)	73.9
ATSS [61]	31.92M	206.86G	71.91	83.62	0.7732	48.1	72.3	53.4	80.29	89.80	0.8478	58.5	84.7	65.0
ATSS†	34.28M	220.80G	77.12	85.56	0.8112	54.1	78.5(+6.2)	61.6	83.45	93.52	88.20	65.5	88.5(+3.8)	74.0
Autoassign [90]	36.00M	203.09G	74.53	84.15	0.7905	47.7	76.6	53.2	81.76	91.46	0.8634	60.4	86.8	67.1
Autoassign†	34.72M	220.96G	77.02	86.19	0.8135	53.0	79.3(+2.7)	59.5	83.81	92.50	0.8794	64.4	88.8(+2.0)	73.0
GFL [91]	32.07M	210.02G	75.07	83.74	0.7919	51.5	76.0	58.2	81.34	90.85	0.8583	63.0	86.3	70.3
GFL†	34.43M	223.97G	77.48	85.64	0.8136	55.8	78.8(+2.8)	64.5	83.79	93.18	0.8824	67.6	89.3(+3.0)	75.9
Guided Anchor [92]	37.23M	206.90G	71.49	82.88	0.7676	45.2	74.1	49.5	78.96	89.70	0.8399	50.0	82.8	55.0
Guided Anchor†	35.90M	219.90G	76.83	86.15	0.8122	53.2	79.5(+5.4)	60.9	82.55	91.61	0.8684	58.8	86.5(+3.7)	66.2
Dynamic rcnn [93]	41.19M	211.35G	74.70	81.92	0.7814	51.7	76.8	59.9	78.19	84.45	0.8120	61.0	81.7	69.5
Dynamic rcnn†	43.51M	222.08G	78.09	83.75	0.8082	56.7	79.8(+3.0)	66.7	82.40	85.32	0.8383	64.2	84.4(+2.7)	73.6
Mask rcnn [94]	43.82M	263.12G	73.75	81.74	0.7754	48.1	75.7	54.7	80.45	86.71	0.8346	61.2	83.5	69.4
Mask rcnn†	46.12M	273.84G	75.93	83.60	0.7958	52.3	79.2(+3.5)	60.2	82.57	86.88	0.8467	63.8	85.1(+1.6)	73.7
Cascade rcnn [62]	68.97M	239.13G	71.59	80.75	0.7589	47.0	73.3	52.0	80.10	85.07	0.8251	62.3	83.0	70.1
Cascade rcnn†	71.28M	249.85G	78.64	83.60	0.8104	56.9	80.2(+6.9)	65.1	82.85	87.00	0.8487	67.2	85.5(+2.5)	77.1
Sparse rcnn [95]	105.97M	153.31G	74.83	81.94	0.7822	50.7	73.9	56.9	77.44	85.90	0.8145	57.1	79.2	65.8
Sparse rcnn†	108.28M	164.04G	77.65	82.90	0.8019	55.1	77.2(+3.3)	62.7	79.17	87.62	0.8318	61.2	83.1(+3.9)	70.1
Libra rcnn [96]	36.66M	216.54G	77.21	84.77	0.8081	54.4	77.9	61.9	79.73	90.17	0.8463	59.0	83.9	64.9
Libra rcnn†	35.33M	229.54G	79.53	86.08	0.8268	58.5	80.7(+2.8)	66.4	83.30	93.84	0.8826	64.5	89.0(+5.1)	71.9

TABLE VII

ABLATION EXPERIMENTS FOR OUR WTRM AND CLSMHA USING THE RETINANET DETECTOR ON DOTA-V1.0 AND HRSID DATASETS IN THE RETINANET [67] DETECTOR. WE VALIDATE THE EFFECTIVENESS OF OUR WTRM AND CLSMHA MODULES IN WTHA-VIT, AS WELL AS CONDUCT ABLATION EXPERIMENTS ON WHETHER WAVELET INVERSE TRANSFORM IS PERFORMED IN WTRM AND FUSED INTO THE FINAL OUTPUT

Model	CLSMHA	WTRM	idwt	Params	GFLOPs	DOTA-V1.0						HRSID					
						P	R	F_1	AP	AP_{50}	AP_{75}	P	R	F_1	AP	AP_{50}	AP_{75}
WTHA-ViT-S (ours)	✗	✗	✗	33.52M	204.11G	71.64	82.58	0.7672	47.5	72.3	52.3	78.83	90.80	0.8439	56.8	81.5	63.8
	✗	✓	✗	34.04M	211.94G	74.58	83.97	0.7900	50.7	74.7	56.6	81.22	92.03	0.8629	60.8	85.4	68.1
	✗	✓	✓	34.53M	215.70G	74.40	83.98	0.7890	50.2	75.0	56.4	81.56	92.38	0.8663	61.0	85.8	67.9
	✓	✗	✗	34.06M	216.88G	73.94	83.70	0.7852	50.1	74.6	56.2	81.11	91.89	86.16	60.9	85.5	68.1
	✓	✓	✗	34.58M	224.71G	74.80	84.03	0.7915	50.8	76.0	56.7	84.49	92.69	0.8840	65.2	89.0	72.0
	✓	✓	✓	35.07M	228.46G	**75.15**	**84.62**	**0.7962**	**52.4**	**76.5**	**60.5**	**85.23**	**94.29**	**0.8953**	**65.6**	**89.5**	**73.0**



(a) (b)

Fig. 11. Comparison of mAP50 (%) versus GFLOPs and Parameters (M) on DOTA-V1.0 of different backbones using RetinaNet for object detection, where T, S, B, M, and L denote models with tiny, small, base, medium, and large size. (a) mAP50 (%) versus GFLOPs. (b) mAP50 (%) versus Parameters (M).

in Table IV, we perform detailed comparison experiments with state-of-the-art algorithms in two representative detectors, i.e., ATSS [61] and Cascade RCNN [62] in mmdetection [63]. Our algorithm has always worked optimally, and the generalizability of our algorithm is further verified. The Transformer-based architecture we employ is currently a widely adopted high-performance architecture. We hope to achieve a performance and compute burden balance based on this architecture.

E. Ablation Analysis on WTHA-ViT Architectures

In the ablation experiments, we train the RetinaNet detector using a 3× schedule (36 epochs) to evaluate the effectiveness of the proposed WTRM and CLSMHA modules separately. We conduct ablation experiments on different WTS and threshold selection strategies proposed by WTRM (see Table VIII). Similarly, we perform ablation experiments on the three

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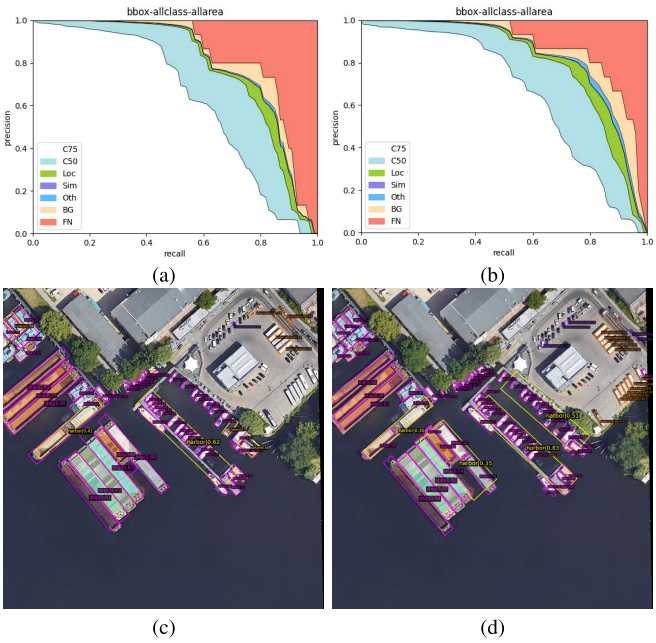

Fig. 12. P–R curves of RVSA [88] and our WTHA-ViT for OOD and the visualization of inference results. (a) P–R curves of RVSA. (b) P–R curves of WTHA-ViT (ours). (c) Inference results of RVSA. (d) Inference results of WTHA-ViT (ours).

head model methods proposed by CLSMHA (see Table IX). Additionally, we conduct a detailed ablation study on the modules and inverse wavelet transform in the whole WTHA-ViT (see Table VII) to assess their individual effectiveness.

1) In the proposed WTRM, we investigate two methods for selecting frequency subbands in the wavelet tree structure for WTS. We evaluate the impact of each method on the overall performance of WTHA-ViT in object detection on the DOTA-V1.0 and HRSID datasets (see Table VIII). Notably, all the frameworks used here are WTHA-ViT-S, with the exception of the method used in WTS. Other settings remain exactly the same to ensure a fair comparison.

The All method involves selecting all subbands obtained from wavelet decomposition based on a threshold, while the Root method selects subbands along the tree structure, discarding both a subband and its child nodes if it is discarded. The results show that the Root method offers better interpretability and leads to improved network performance, achieving an mAP of 89.5% on the HRSID and an mAP of 76.5% on the DOTA-V1.0.

2) In WTRM, the retention of each subband in the wavelet tree structure depends on the relationship between its wavelet coefficients and the threshold T. We propose three strategies for selecting the threshold T to evaluate their impact on WTHA-ViT (see Table VIII).

Specifically, the Fixed strategy uses a manually designed fixed threshold value for each level of wavelet decomposition. The EZW strategy applies a threshold selection method inspired by EZW coding principles. The Self strategy utilizes statistical characteristics of wavelet coefficients at each decomposition level as the threshold T. Results indicate that the EZW and Self methods achieve better network performance, with an mAP of 89.5% on the HRSID and an mAP of 76.5% on the DOTA-V1.0.

TABLE VIII

WTS AND THRESHOLD T SELECTION ABLATION EXPERIMENTS OF WTRM ON HRSID AND DOTA DATASETS IN THE RETINANET [67] DETECTOR

Model	WTS		Threshold			mAP_{50}	
	All	*Root*	*Fixed*	*EZW*	*Self*	HRSID	DOTA
WTHA-ViT	✓					89.0	75.9
		✓				**89.5**	**76.5**
			✓			89.0	76.3
				✓		89.4	76.1
					✓	**89.5**	**76.5**

TABLE IX

HEAD MODEL ABLATION EXPERIMENTS OF CLSMHA ON HRSID AND DOTA DATASETS IN THE RETINANET [67] DETECTOR

Model	Head Model			mAP_{50}	
	Hand	*Network*	*Similarity*	HRSID	DOTA
WTHA-ViT	✓			89.0	**76.5**
		✓		**89.5**	76.4
			✓	89.3	76.1

3) In CLSMHA, after applying the CLS to the input features to obtain channel-wise distributed high-frequency and low-frequency features, it is important to model the head importance to align them with the heads of multihead attention in the Transformer encoder. Considering the frequency bias of deep learning neural networks, we propose three methods to model the head importance and examine their impact on the overall performance of WTHA-ViT, as shown in Table IX.

The Hand method assigns higher weights to the channels of low-frequency features and lower weights to the channels of high-frequency features using a manual design. The Network method learns head importance through the network, focusing attention on different channel frequency features. The Similarity method models head importance based on the correlation between channels, allowing for the representation of different frequency feature importance in different channels. From the results in Table IX, it can be observed that the Network and Hand methods achieve better network performance. They attain an mAP of 89.5% on the HRSID and an mAP of 76.5% on the DOTA-V1.0.

4) Finally, we validate the effectiveness of our WTRM and CLSMHA modules in WTHA-ViT and conduct ablation experiments on wavelet inverse transform and fusion in the final output. The experiments are performed on the DOTA-V1.0 dataset using the Retinanet detector, and the results are summarized in Table VII.

The results in Table VII demonstrate that our proposed WTRM and CLSMHA modules significantly improve remote sensing object detection performance. The best performance is achieved when wavelet inverse transform is performed and fused into the final output, with an mAP of 76.5% on the DOTA-V1.0 and 89.5% on the HRSID.

F. Visualization of Learned Visual Representation

1) As demonstrated in related work, PVT [22] employs downsampling learning on the K and V features through

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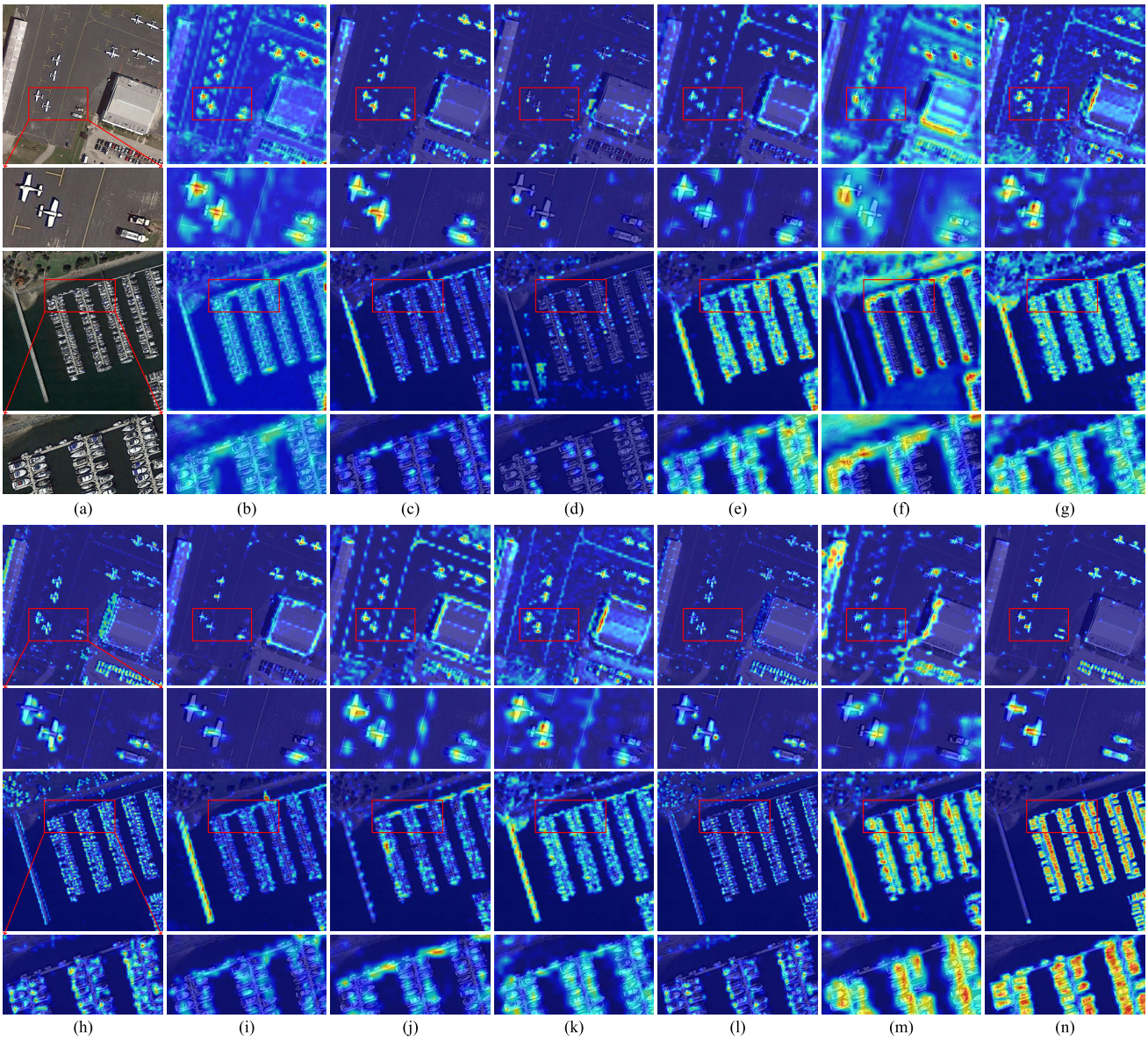

Fig. 13. Visualization of CAM for different models on DOTA-V1.0 dataset. (a) Original images. (b) ResNet50 [55]. (c) PVT-T [22]. (d) Swin-T [13]. (e) Twins-SVT-S [21]. (f) Wavevit-S [23]. (g) LIT-T [58]. (h) DAT++-T [57]. (i) UniFormer-S [56]. (j) GFNet-T [27]. (k) AFNO-T [28]. (l) SpectFormer-S [26]. (m) RVSA [88]. (n) WTHA-ViT-S (ours).

pooling operation, while Wavevit [23] utilizes a simple wavelet transform and concatenates the four frequency components along the channel dimension. As can be seen from Fig. 3, we compare and analyze our method in three images from natural image dataset (ImageNet [64]) and remote sensing image dataset (DOTA [37]). It can be observed that when the image is downsampled using the pooling method and then restored using bilinear interpolation, the reconstructed images (b) are blurred and cannot capture the specific structure of the object. The reconstructed images (c) can only preserve the main features of the object due to the discarded high-frequency detail subbands. In contrast to the previous two methods, our wavelet tree frequency selection method reconstructs the images (d) that not only preserve the main features of the object but also express the internal contours of the object in detail. In addition, the frequency features it retains are adaptive, and different frequency selection processes are carried out according to different images.

2) In order to provide further insights into the visual representations learned by WTHA-ViT and gain a better understanding of the model's performance, class activation maps (CAM) are generated to locate discriminative regions in the image and display the prominent structures recognized by the model. For the DOTA-V1.0 dataset, we selected three images from various scenes and generated CAMs using the Retinanet detector, as shown in Fig. 13. Particularly in dense scenes, our WTHA-ViT-S demonstrates significantly enhanced attention concentration around the object in comparison to other advanced algorithms, thus displaying its superiority. Notably, while the original network allocates considerable attention to regions surrounding the object, it also devotes attention to nonobject areas, thereby introducing interference.

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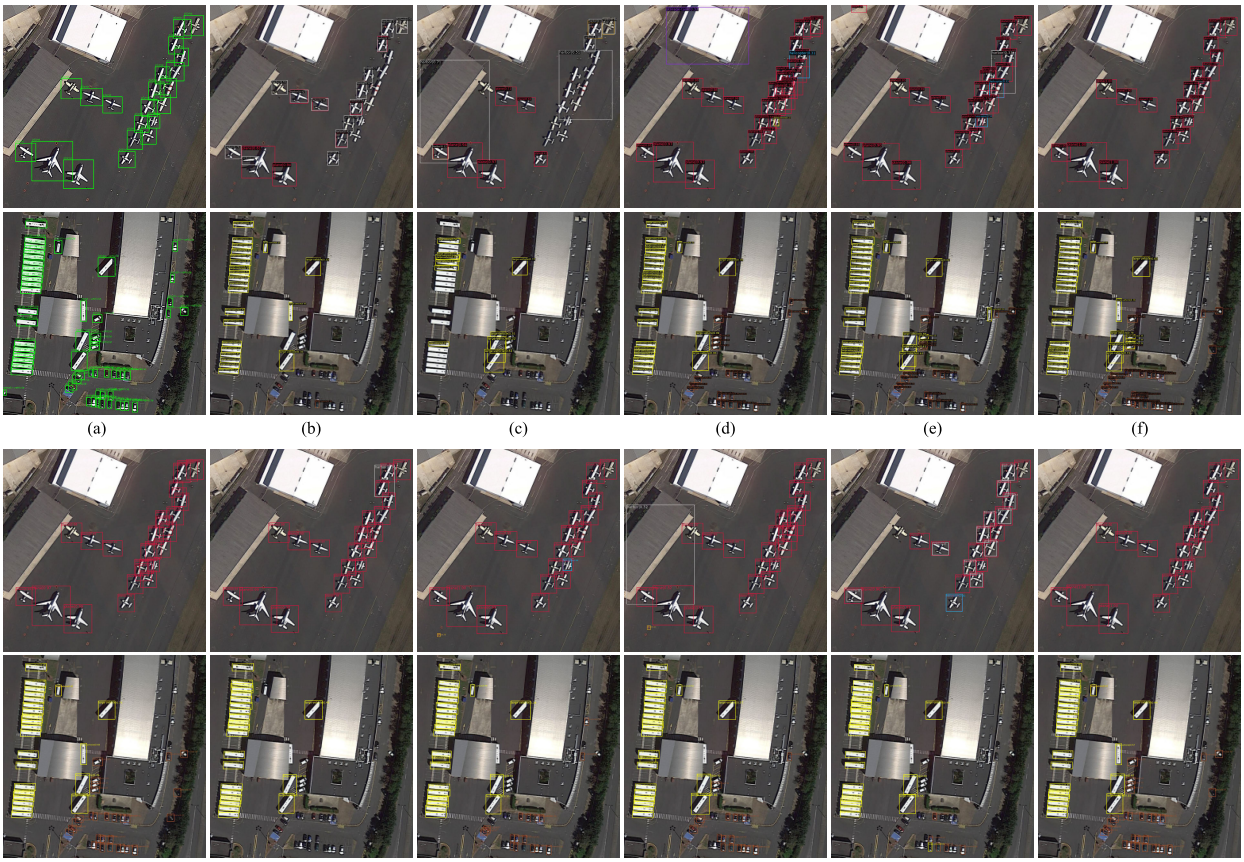

Fig. 14. Visualization of robustness for different models on DOTA-V1.0 dataset. (a) Original images with GT. (b) PVT-T [22]. (c) Swin-T [13]. (d) Twins-S [21]. (e) Wavevit-S [23]. (f) LIT-T [58]. (g) DAT++-T [57]. (h) UniFormer-S [56]. (i) GFNet-T [27]. (j) AFNO-T [28]. (k) SpectFormer-S [26]. (l) WTHA-ViT-S (ours).

TABLE X

ROBUSTNESS VERIFICATION (%) OF NOISY DATASET ON DOTA-V1.0 DATASET FOR DIFFERENT MODELS. † REPRESENTS THE INFERENCE RESULT OF THE MODEL ON THE ORIGINAL NOISE-FREE DATASET

Backbone	AP	AP_{50}	AP_{75}	AP_{50}†	ΔAP_{50}
ResNet50 [55]	12.2	24.5	10.5	70.2	-65.1%
PVT-T [22]	18.2	33.5	17.8	73.4	-54.4%
Swin-T [13]	20.7	35.9	19.2	74.9	-52.1%
Twins-SVT-S [21]	30.5	52.3	30.8	73.0	-28.4%
WaveViT-S [23]	28.0	48.0	27.7	74.5	-35.6%
LIT-T [58]	29.3	49.8	28.7	74.1	-32.8%
DAT++-T [57]	40.2	64.5	44.8	74.3	-13.2%
UniFormer-S [56]	25.4	43.8	25.3	74.3	-41.0%
GFNet-T [27]	36.4	59.3	39.1	72.4	-18.1%
AFNO-T [28]	39.4	62.7	42.3	74.6	-16.0%
SpectFormer-S [26]	26.8	45.5	27.0	73.4	-38.0%
WTHA-ViT-S (**ours**)	**44.7**	**68.8**	**48.8**	**76.5**	**-10.1%**

In contrast, after replacing with WTHA-ViT-S, the network's ability to focus attention around the object is further improved, and attention to nonobject areas is effectively suppressed. As shown in Fig. 13(n), our WTHA-ViT-S focuses the attention of the network on aircraft and ship objects.

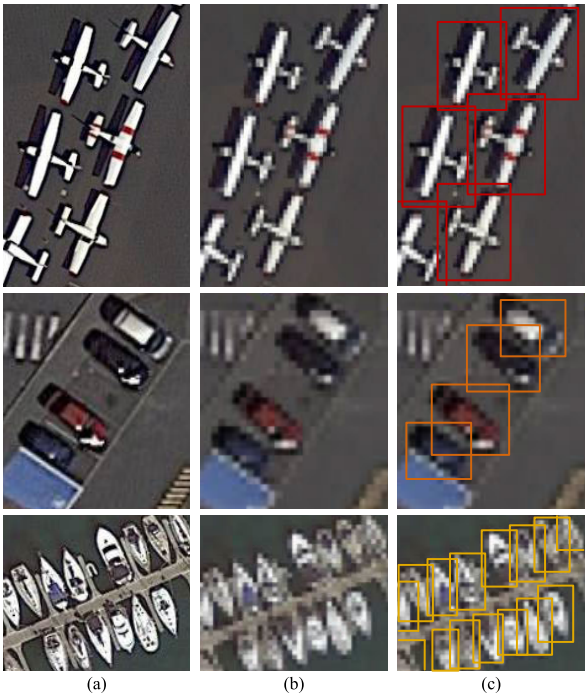

Fig. 15. Details of the detection results of our model for pixelate images. (a) Original images. (b) Pixelate the original images. (c) Inference results of WTHA-ViT-S (ours).

G. Robustness Verification of Noisy Datasets

Benefiting from the capabilities of wavelet decomposition and reconstruction [97], wavelets have found primary applications in image denoising [29] and compression [98]. Leveraging the wavelet tree structure, our WTHA-ViT can

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TABLE XI

QUANTITATIVE EXPERIMENT OF MODEL TRANSFERABILITY. THE BASELINE IS RETINANET [67] AND THE MODELS TRAINED WITH DOTA-V1.0 ARE TRANSFERRED TO TWO LARGE-SCALE DATASETS DIOR AND LEVIR. THE GENERALIZATION PERFORMANCE OF OUR METHOD IS COMPARED WITH ADVANCED METHODS. THE RED FONT REPRESENTS THE BEST AP_{50}, AND THE BLUE FONT REPRESENTS THE SECOND BEST

Backbone	Params	GFLOPs	DIOR										LEVIR		
			PL	BD	BR	GTF	VE	SH	TC	BC	ST	HA	PL	SH	ST
ResNet50 [55]	36.39M	215.46G	20.2	35.6	4.3	26.1	9.7	32.7	65.1	35.2	31.1	14.4	20.5	9.5	41.0
PVT-T [22]	21.62M	158.11G	26.4	37.5	5.8	41.6	12.3	36.2	64.1	50.5	33.0	20.8	24.6	10.9	50.9
Swin-T [13]	37.11M	221.64G	20.6	33.5	7.3	35.5	11.7	32.6	65.9	49.0	33.7	20.8	19.7	12.0	45.6
Twins-SVT-S [21]	32.44M	185.68G	21.4	41.3	5.5	39.5	11.7	34.9	66.6	47.8	32.3	21.9	20.3	12.2	50.9
UniFormer-S [56]	29.93M	199.68G	22.3	36.9	5.1	37.6	10.7	36.2	64.2	32.1	29.0	21.4	21.3	10.7	44.3
DAT++-T [57]	32.34M	211.67G	20.6	35.8	5.9	34.8	9.6	35.2	63.3	34.7	28.0	22.4	18.9	11.0	41.1
LIT-T [58]	28.27M	196.00G	20.1	32.3	4.0	30.5	7.7	34.3	56.9	18.2	27.8	18.9	19.2	8.9	38.2
WaveViT-S [23]	28.11M	216.51G	22.9	36.1	4.6	37.6	11.6	34.6	66.3	54.5	31.7	24.4	16.6	10.7	47.0
SpectFormer-S [26]	50.22M	206.92G	24.1	34.3	5.0	37.0	11.6	36.4	67.5	50.0	33.4	25.4	17.3	10.6	46.9
GFNet-T [27]	54.29M	178.82G	17.3	32.5	4.9	33.3	8.2	33.2	60.1	24.5	24.2	20.6	18.3	10.6	38.5
AFNO-T [28]	23.51M	178.82G	20.8	35.5	6.1	34.6	9.9	34.8	63.2	34.2	27.8	23.2	18.8	10.8	41.1
WTHA-ViT-S(**ours**)	35.07M	228.46G	25.6	38.0	6.1	42.2	13.9	37.6	70.2	61.6	34.8	25.3	20.5	11.9	50.8

PL: plane. BD: baseball diamond. BR: bridge. GTF: ground track field. VE: vehicle. SH: ship. TC: tennis court. BC: baseball court. ST: storage tank. HA: harbor.

TABLE XII

COMPARISON OF THE NUMBER OF TYPES, THE NUMBER OF IMAGES, THE NUMBER OF INSTANCES, AND THE YEAR OF PUBLICATION OF THE THREE OPTICAL REMOTE SENSING OBJECT DETECTION DATASETS

	Categories	Images	Instances	Year
DOTA [37]	15	2806	188282	2017
DIOR [59]	20	23463	192472	2018
LEVIR [60]	3	21952	11028	2018

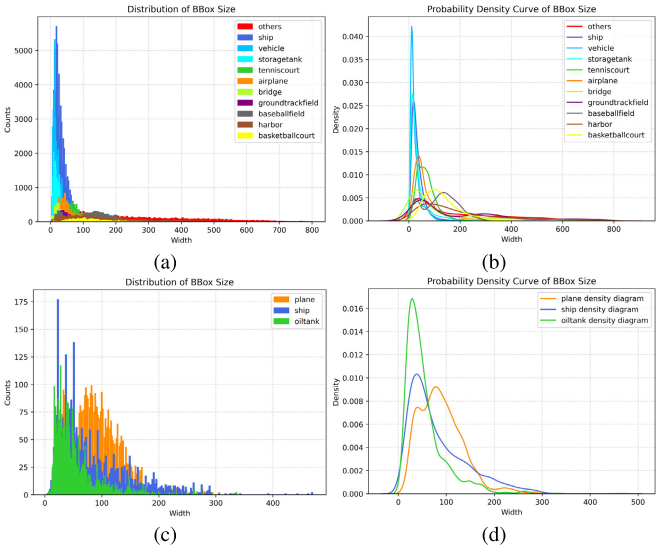

Fig. 16. Histogram of instances bounding box size distribution and probability density curve diagram of DIOR [59] and LEVIR [60] datasets. (a) Distribution of bounding box size in DIOR. (b) Probability density curve in DIOR. (c) Distribution of bounding box size in LEVIR. (d) Probability density curve in LEVIR.

effectively fuse features of different frequencies, making it well-suited for handling noisy images. So, we preprocess the test set of the DOTA-V1.0 dataset by pixelating the images and then employ various models with similar performance on the original testset to infer on this noisy test set, aiming to validate their robustness. The quantitative analysis in Table X reveals that when comparing the detection results of the noisy test set with the original dataset, our WTHA-ViT-S model exhibits the smallest performance drop, with only an 10.1% decrease in performance. The visualization of the detection results in Figs. 14 and 15 demonstrates that our WTHA-ViT adapts well to noisy images compared to other state-of-the-art models. Fig. 15(b) provides local enlarged details of the pixelated images, and Fig. 15(c) showcases their corresponding inference results. Through experiments, we have qualitatively and quantitatively verified that our WTHA-ViT possesses excellent robustness, especially in processing noisy images.

H. Exploration of Transferability and Generalization Ability

Further research on the transferability of the network model and its generalization ability on larger scale datasets is helpful to our method, which has practical application value. He et al. [99] is a pioneering work, which is the first attempt of super-resolution mapping (SRM) model to transfer the model in a large spatial range and a large time range to the greatest extent. According to He et al. [99], we apply the model to the data that we have never seen before in the testing stage, and compare the detection performance on the new datasets, so as to explore the transferability of our model and the generalization ability of the network model. As can be seen from Table XII, DIOR and LEVIR datasets are large-scale on the object categories, on the object instances number, and on the total images number, so we choose them to evaluate the transferability of the model trained on DOTA-V1.0 datasets. Fig. 16 is the distribution histogram and probability density diagram of the sizes of instances of two datasets. The figure

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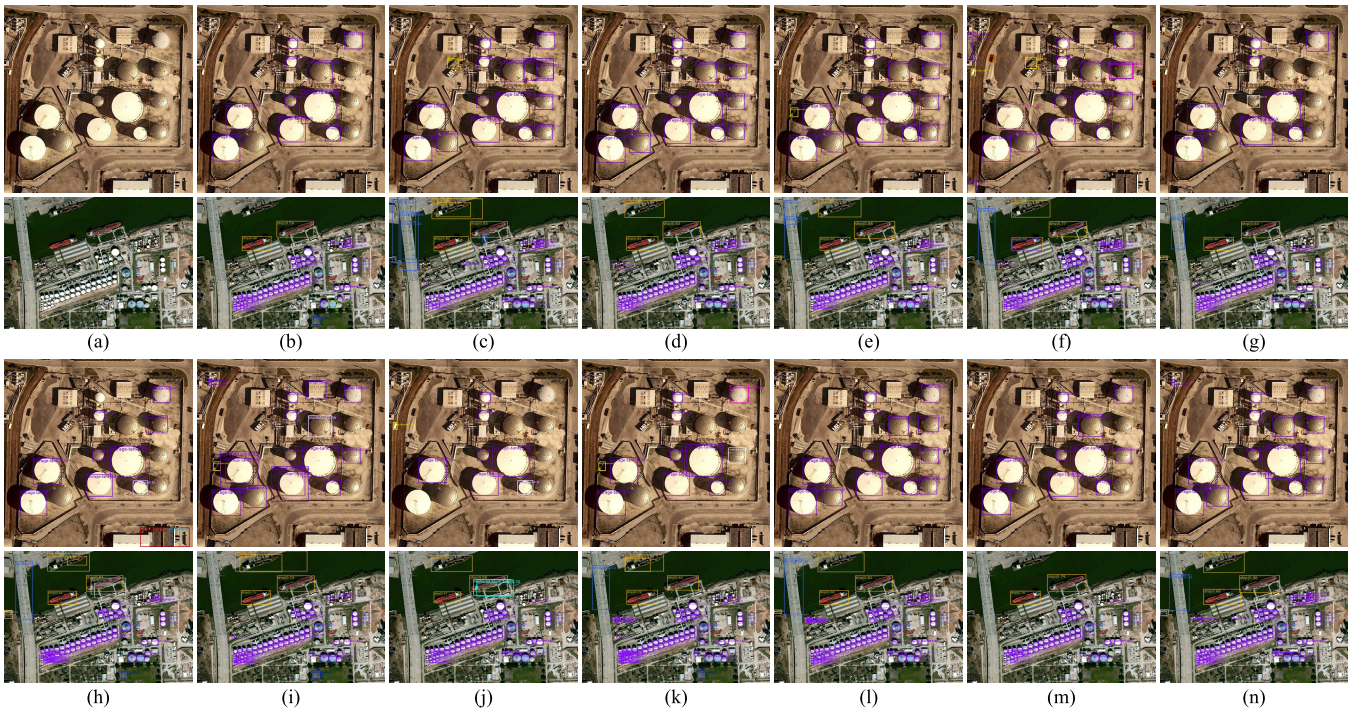

Fig. 17. Visualization of the results of transferability experiment. The first line are the inference results of DIOR dataset, and the second line are the inference results of LEVIR dataset. (a) Original images. (b) ResNet50 [55]. (c) PVT-T [22]. (d) Swin-T [13]. (e) Twins-SVT-S [21]. (f) Wavevit-S [23]. (g) LIT-T [58]. (h) DAT++-T [57]. (i) UniFormer-S [56]. (j) GFNet-T [27]. (k) AFNO-T [28]. (l) SpectFormer-S [26]. (m) RVSA [88]. (n) WTHA-ViT-S (ours).

shows that the two datasets have a large range of object size variations, not only in terms of spatial resolutions, but also in the aspect of inter- and intra-category size variability across objects. It should be noted that these two datasets may have categories that are not available in DOTA, so we only evaluate the AP of the categories that they both have.

The generalization performance of our method is compared with that of the advanced method, as shown in Table XI. Since the annotated categories of these two datasets are not exactly the same as those in DOTA-V1.0, we report separately the AP_{50} metrics for the categories they share. It can be quantitatively analyzed that the transferability and generalization ability of our method are excellent, and most of the categories in the comparison method have reached the top-2 performance. Fig. 17 visualizes the inference results of the transferability experiment of the model in Table XI on DIOR and LEVIR datasets. The detection results on the two new datasets more intuitively reflect that our model has better generalization ability than other advanced models. As shown in Fig. 17(n), our WTHA-ViT-S has the least amount of missed detection in the detection of storage tanks.

Theoretically, our WTHA-ViT can be applied to other tasks, such as classification, segmentation, and other downstream tasks. However, because the tasks such as classification and segmentation are inconsistent with the motivation of our paper, we do not carry out research on other tasks in the article. In the prospect, we hope to do in-depth research in other fields in the future.

VI. Conclusion

In this article, we present an innovative and effective framework named WTHA-ViT, which is a Transformer encoder structure based on a wavelet tree structure and CLS. As a versatile backbone replacement for object detection tasks, WTHA-ViT effectively leverages wavelet transformation to enhance frequency attention. The model intelligently selects frequency components while reducing computational complexity by traversing the wavelet decomposition tree structure. It skillfully captures interactions with varying frequencies and long-range dependencies, ultimately converging at the head to model their importance in diverse combinations. Extensive experiments on four object detection datasets confirm WTHA-ViT's superiority over state-of-the-art models, highlighting its efficacy and robustness. Future research will explore WTHA-ViT's application in various visual tasks and investigate its potential in image denoising, enhancement, and reconstruction.

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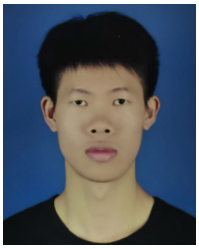

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